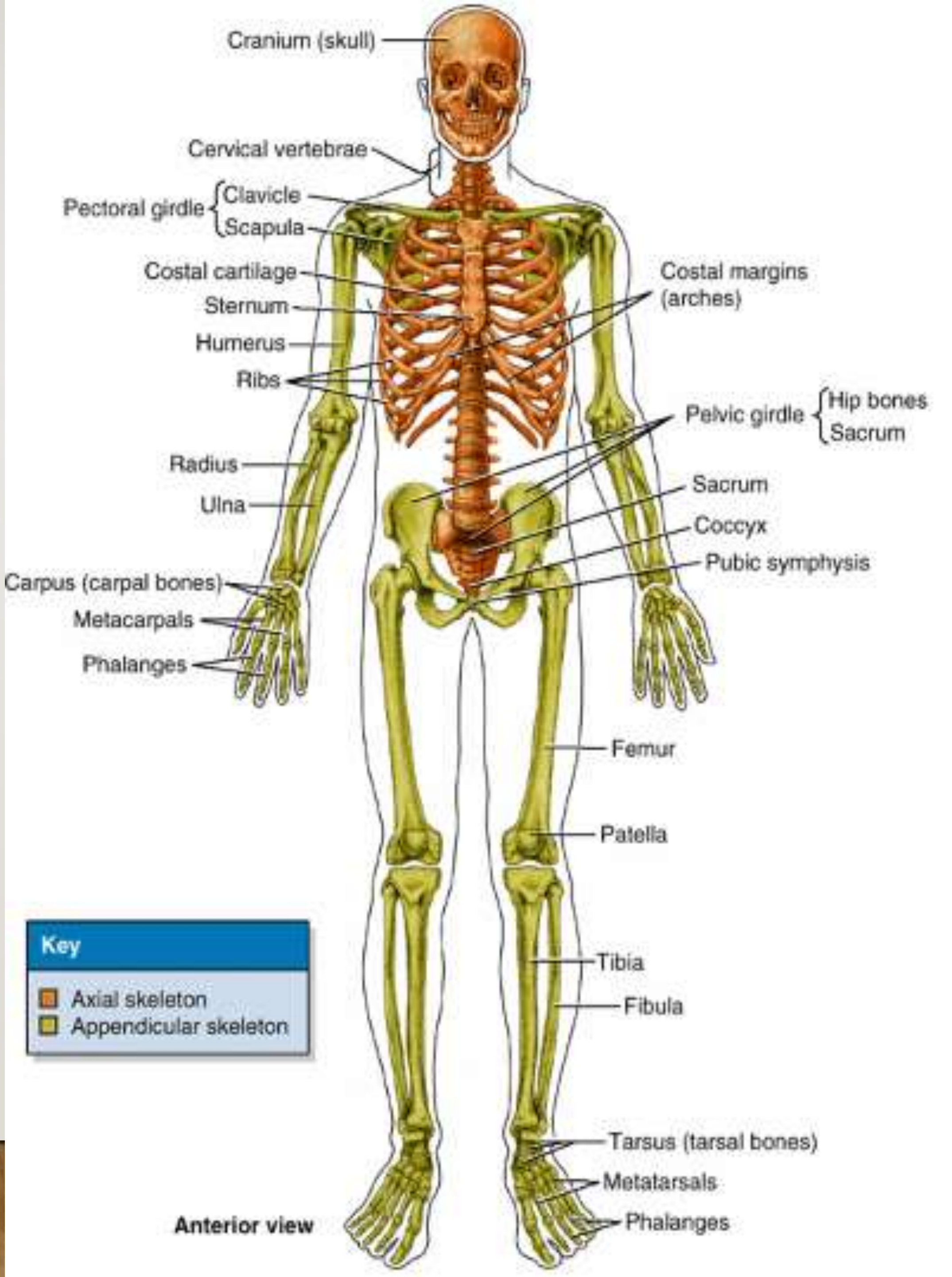
SKELETAL SYSTEM

THE HUMAN SKELETON:

- The human skeleton consists of a series of bones articulated together to form joints.
- **The human skeleton is divided into:**
  1. Axial skeleton
  2. Appendicular skeleton

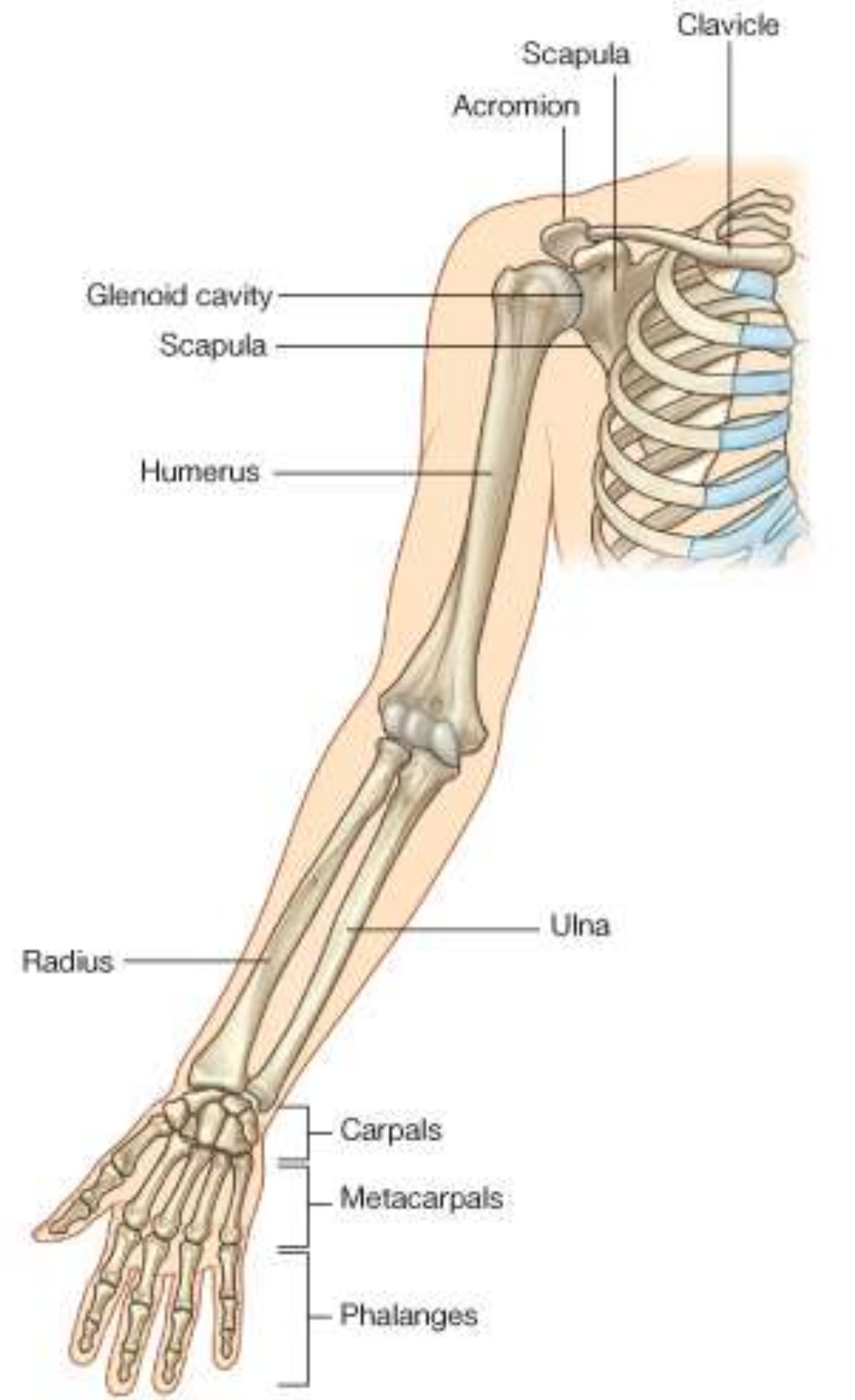


BONES OF UPPER LIMB

The upper limb is composed of the shoulder girdle, arm, forearm and hand.

The bones of these segments are as follows:

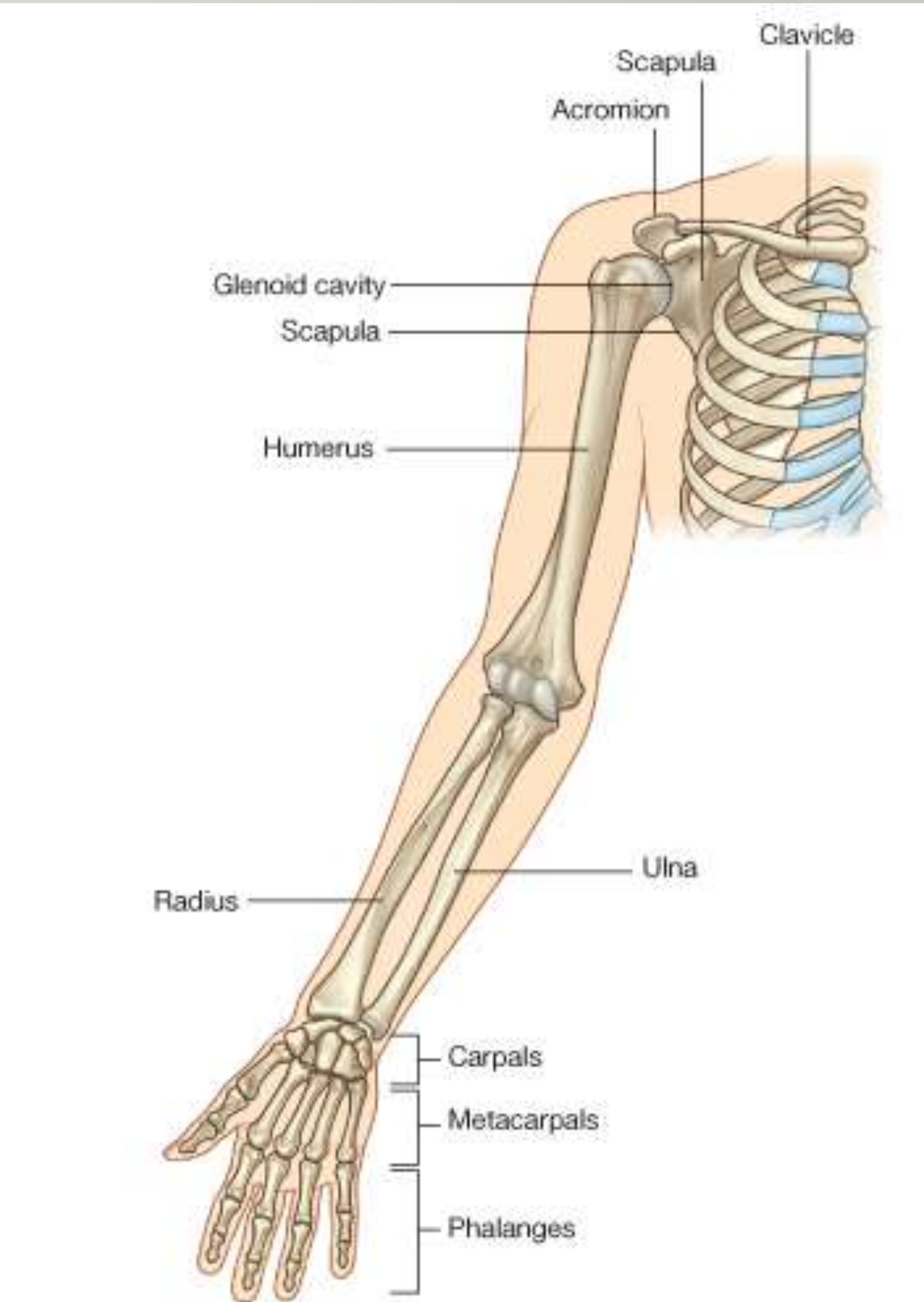
1. Bones of **the shoulder girdle**: are the **scapula and clavicle** together. These 2 bones surround the upper part of the chest like a belt.
2. The bone of the **arm**: is the **humerus**.
3. Bones of the **forearm** are **radius** (laterally), and **ulna** (medially).
4. Bones of the **hand:** are the **carpals, metacarpals and phalanges.**



THE CLAVICLE

Functions:

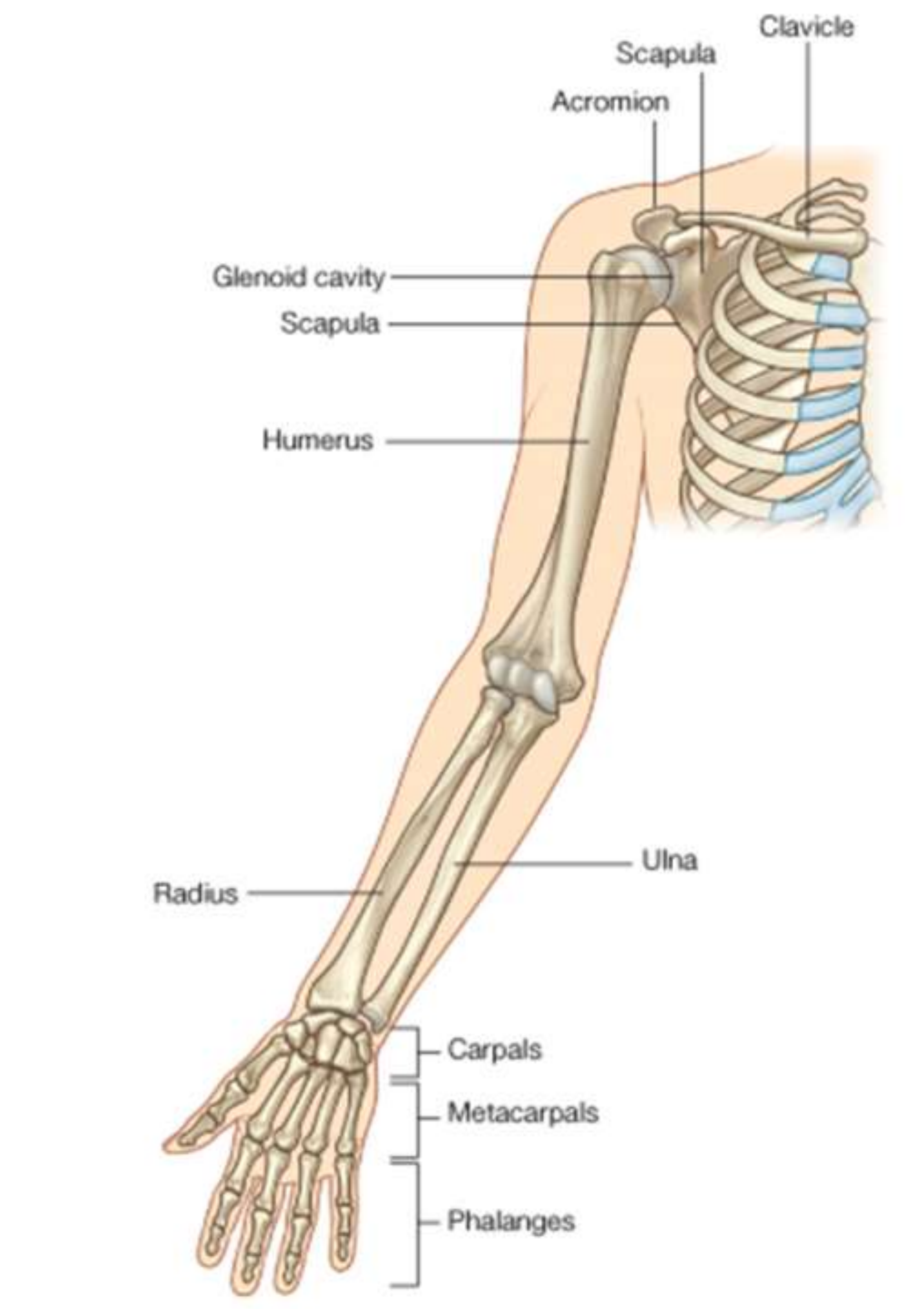
1. It acts as a strut to carry the weight of the limb. It allows the limb to swing away from the trunk.
2. It transmits the weight of the upper limb to the trunk through its articulation with the sternum.

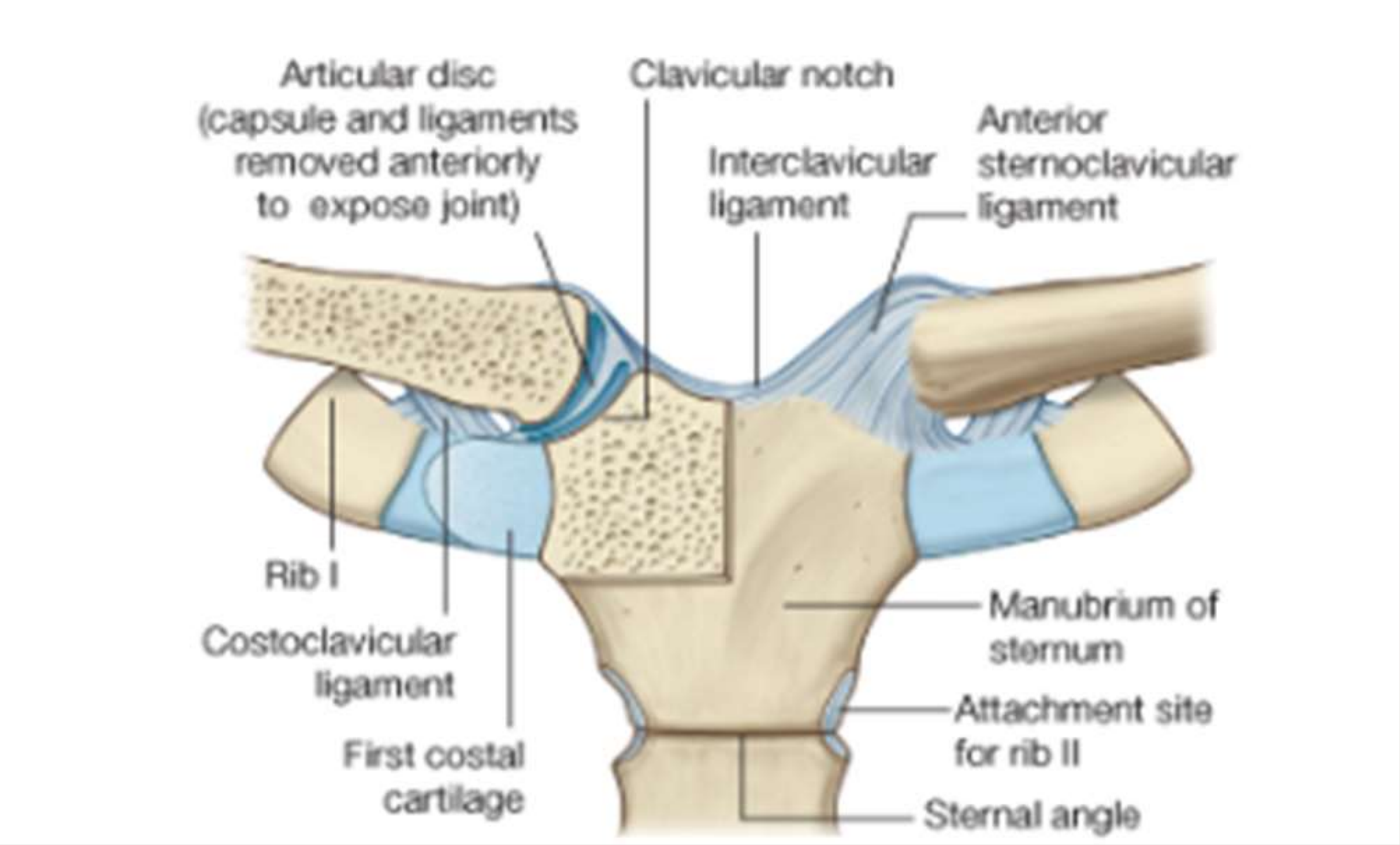


THE CLAVICLE

- It is a **long bone**.
- It has a shaft that lies horizontaly, and two ends medial and lateral.
- **Articulations:**

1. The medial end is thick and rounded and articulates with the clavicular notch of the manubrium sterni to form **sternoclavicular joint**.
2. The lateral end is fattened and articulates with the acromion of the scapula to form the **acromioclavicular joint**.

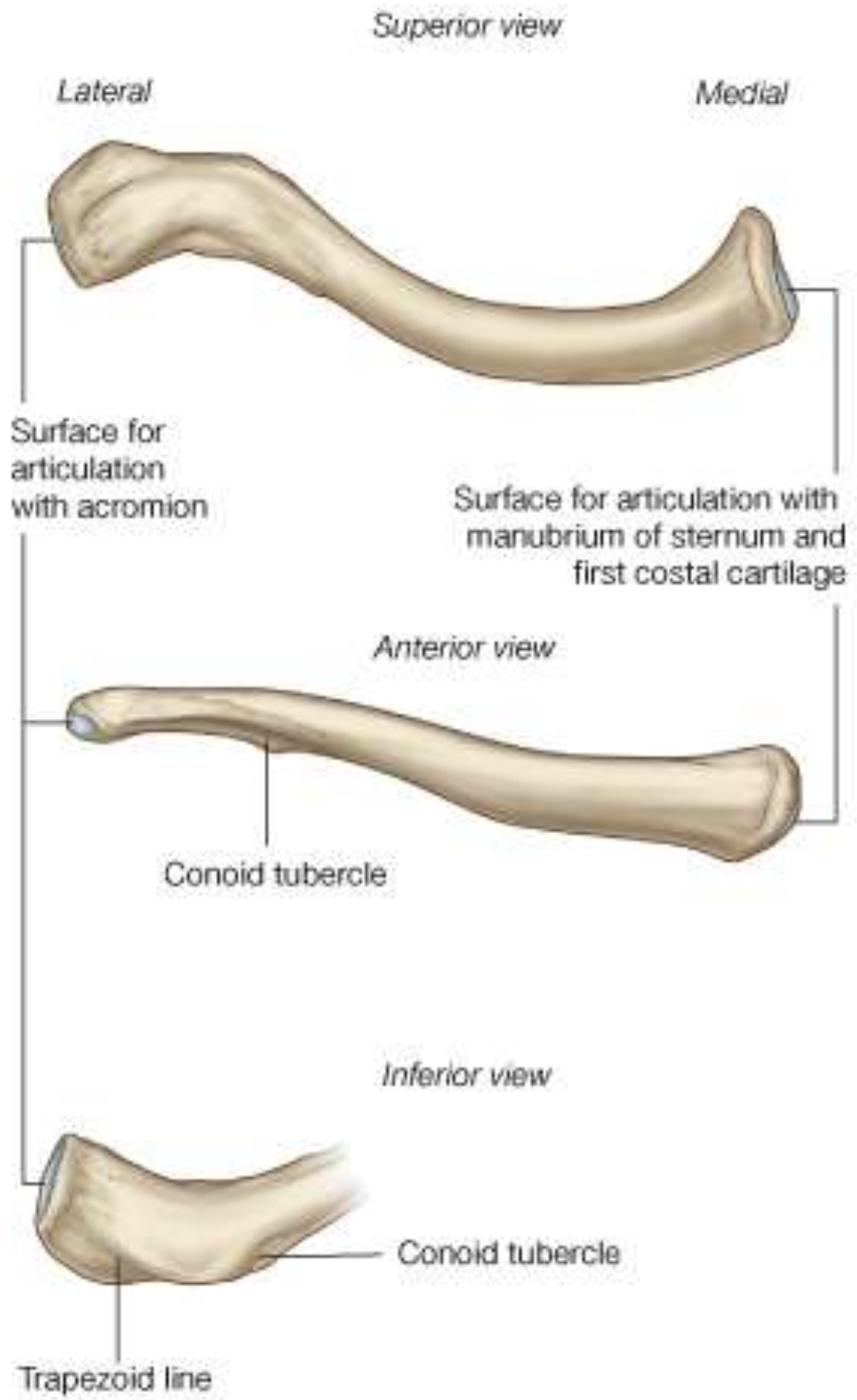




Articular disc
(capsule and ligaments
removed anteriorly
to expose joint)
Clavicular notch
Anterior
sternoclavicular
ligament
Interclavicular
ligament
Rib I
Costoclavicular
ligament
First costal
cartilage
Manubrium of
sternum
Attachment site
for rib II
Sternal angle

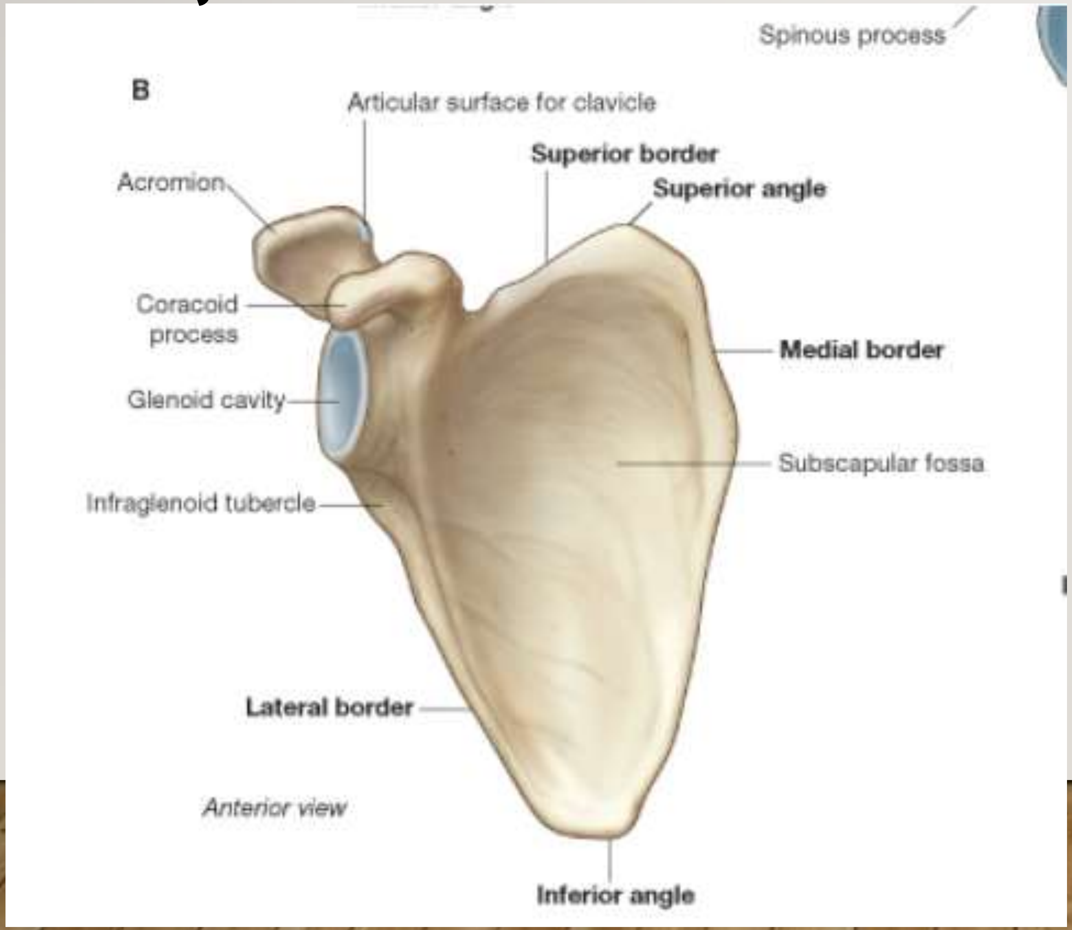
THE CLAVICLE

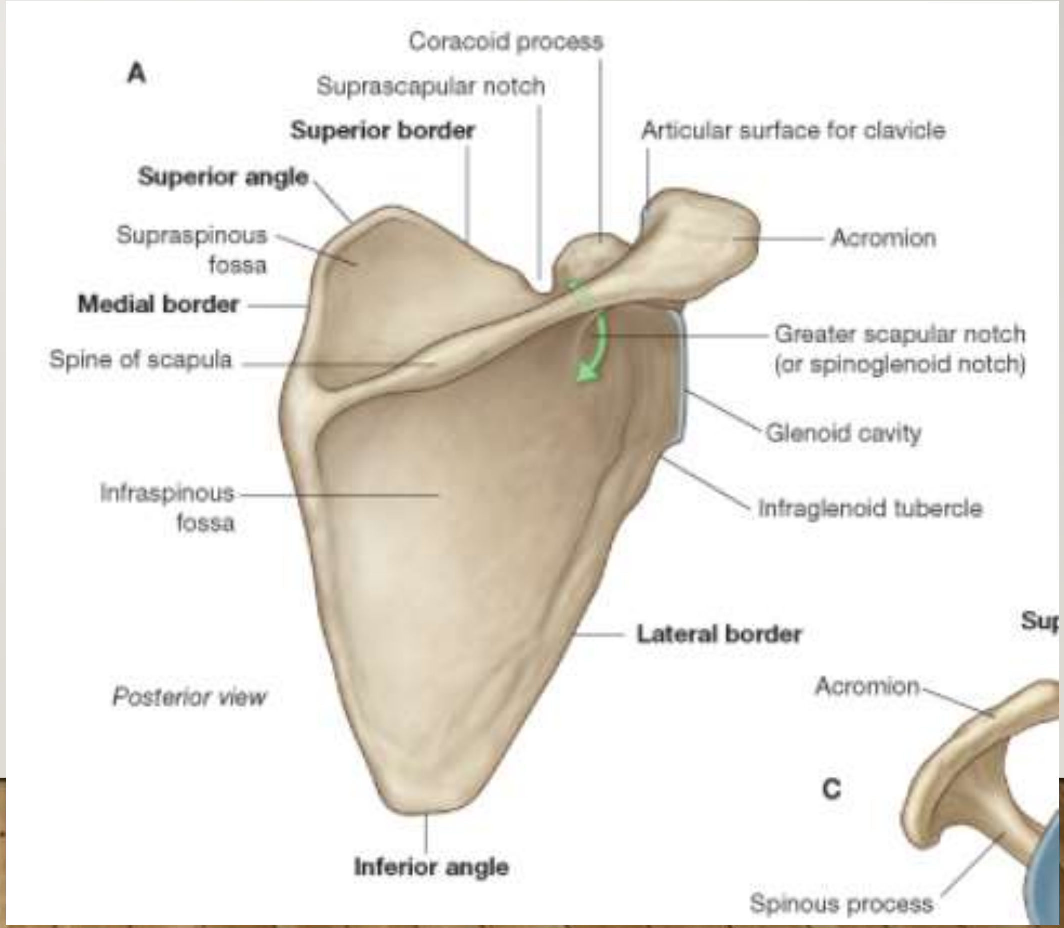
- Its shaft presents a double curvature resembling the letter S.
- Its medial 2/3 is convex forwards while its lateral 1/3 is convex backwards.
- SURFACE LANDMARKS: The clavicle can readily be felt beneath the skin throughout its length (the two ends and the shaft).

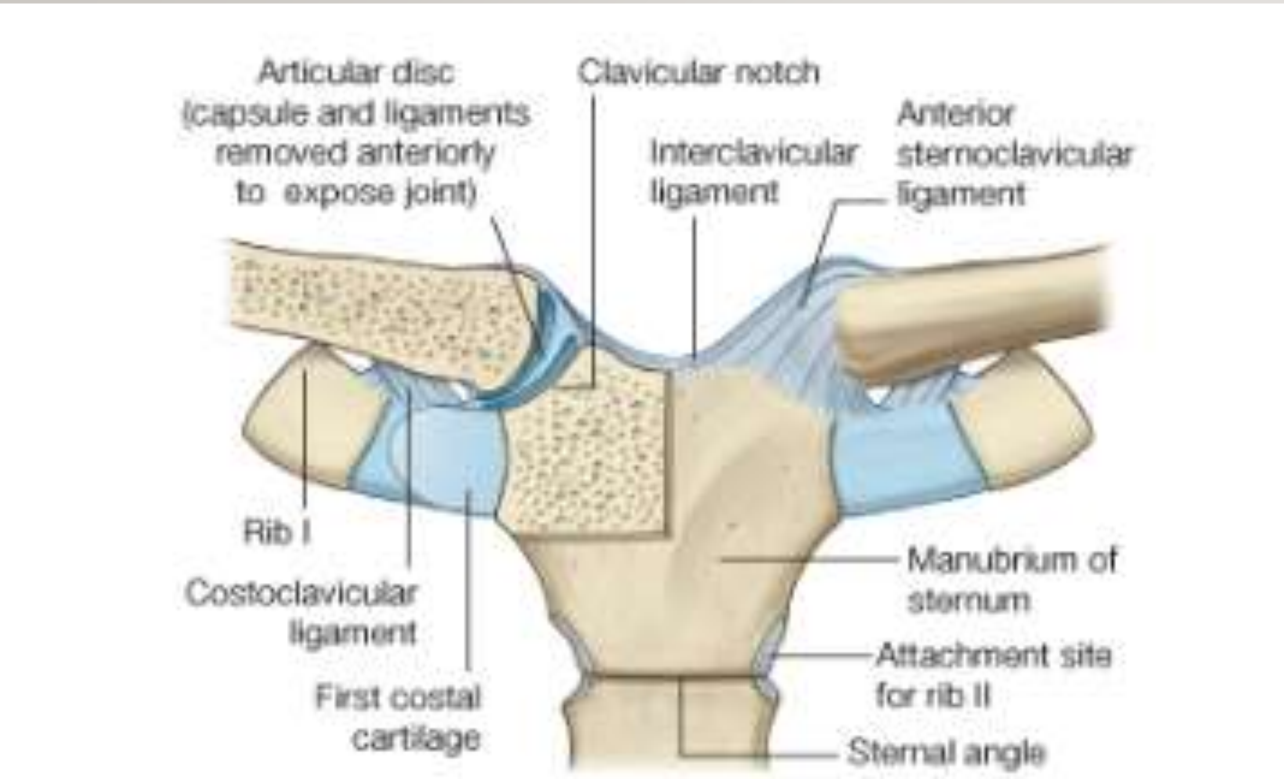


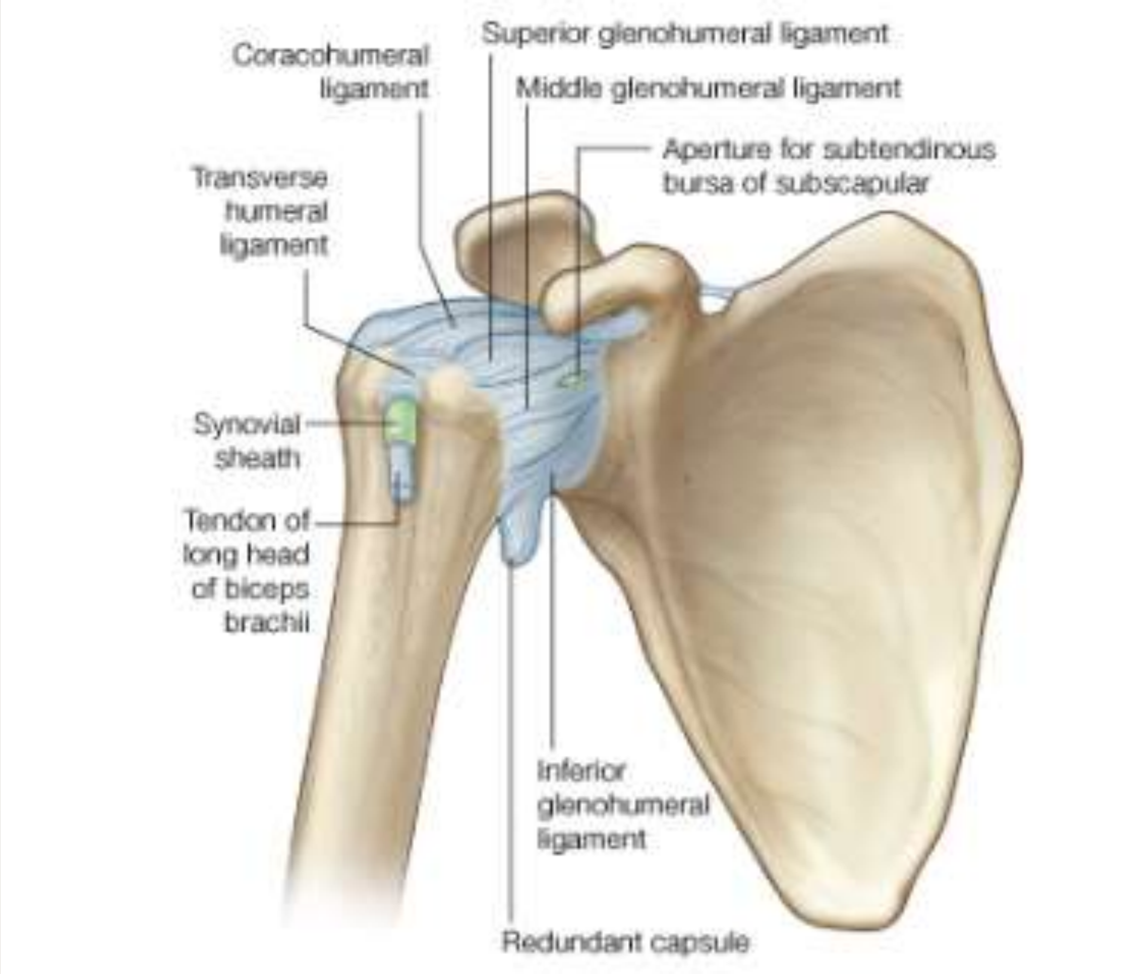
THE SCAPULA

- It is a **flat** triangular in shape having three borders (superior, medial and lateral), two surfaces (anterior and posterior) and three processes (spine, acromion and coracoid process).
- **ARTICULATIONS:** It articulates with the clavicle by its acromion to form the **acromioclavicular joint** and with the head of humerus by the glenoid fossa to form **shoulder joint**.





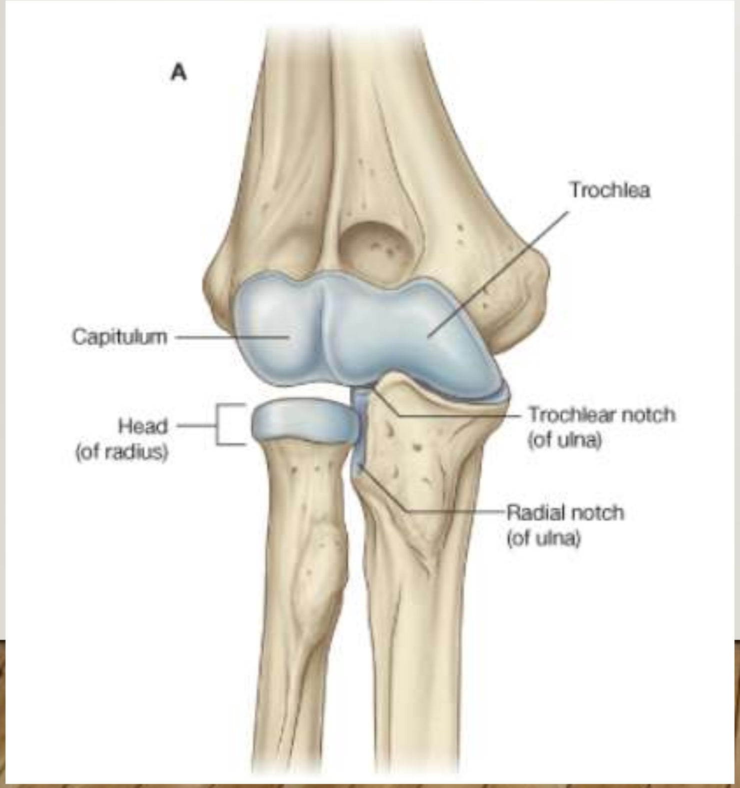

Articular disc
(capsule and ligaments
removed anteriorly
to expose joint)
Clavicular notch
Interclavicular
ligament
Anterior
sternoclavicular
ligament
Rib I
Costoclavicular
ligament
First costal
cartilage
Manubrium of
sternum
Attachment site
for rib II
Sternal angle

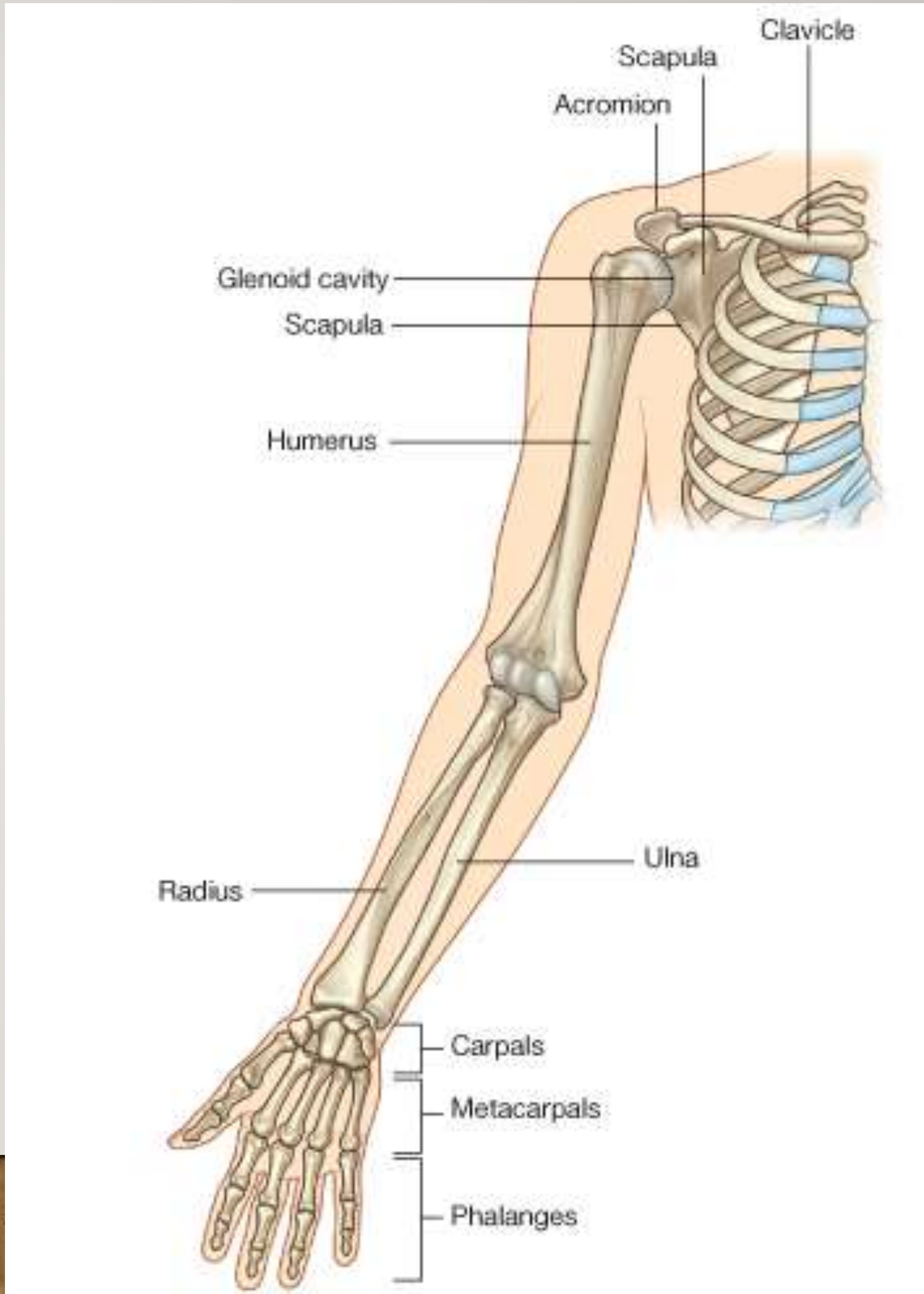

Coracohumeral
ligament
Superior glenohumeral ligament
Middle glenohumeral ligament
Aperture for subtendinous
bursa of subscapular
Transverse
humeral
ligament
Synovial
sheath
Tendon of
long head
of biceps
brachii
Inferior
glenohumeral
ligament
Redundant capsule

THE HUMERUS

- It is a **long bone**.
- It is made up of a shaft, upper and lower ends.
- **Articulations:**

1. With the glenoid cavity of the scapula by its head at the **glenohumeral articulation (shoulder joint).**
2. With the radius by its capitulum and with the ulna by its trochlea at the **elbow joint**.

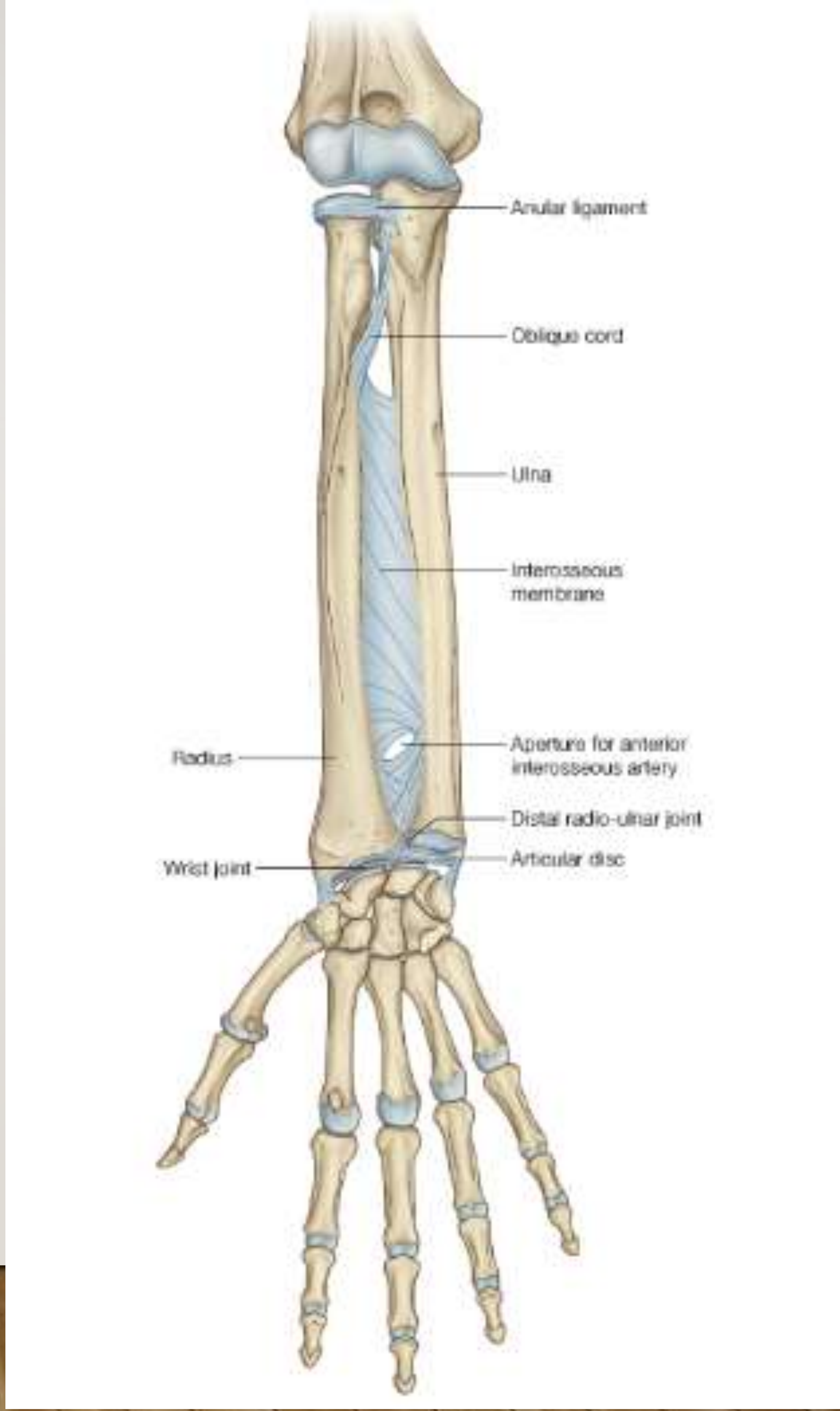


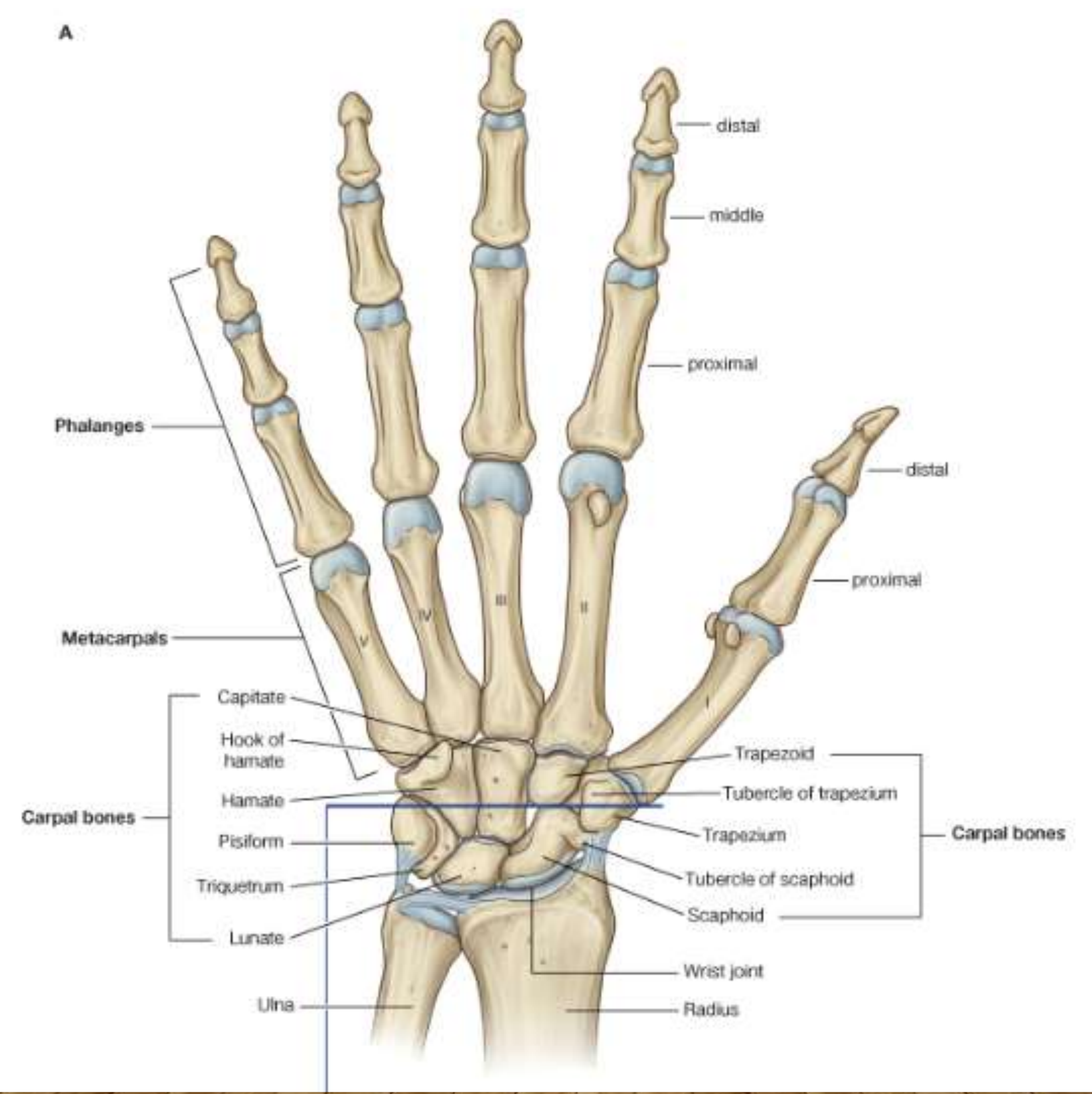


THE RADIUS

- It is the lateral bone of the forearm.
- It is a **long bone** made up of a shaft, an upper end (head) and a lower end.
- **Articulations:**

1. The upper surface of the head of the radius is concave; articulates with the capitulum of the lower end of the humerus (**elbow joint**).
2. The head of the radius articulates medially with the radial notch of the ulna to form the **superior radio-ulnar joint**.
3. The Lower end of the radius articulates medially with head (lower end) of the ulna to form the **inferior radio-ulnar joint**.
4. The Lower end of the radius articulates inferiorly with the carpal bones to form the **wrist (radiocarpal) joint**.

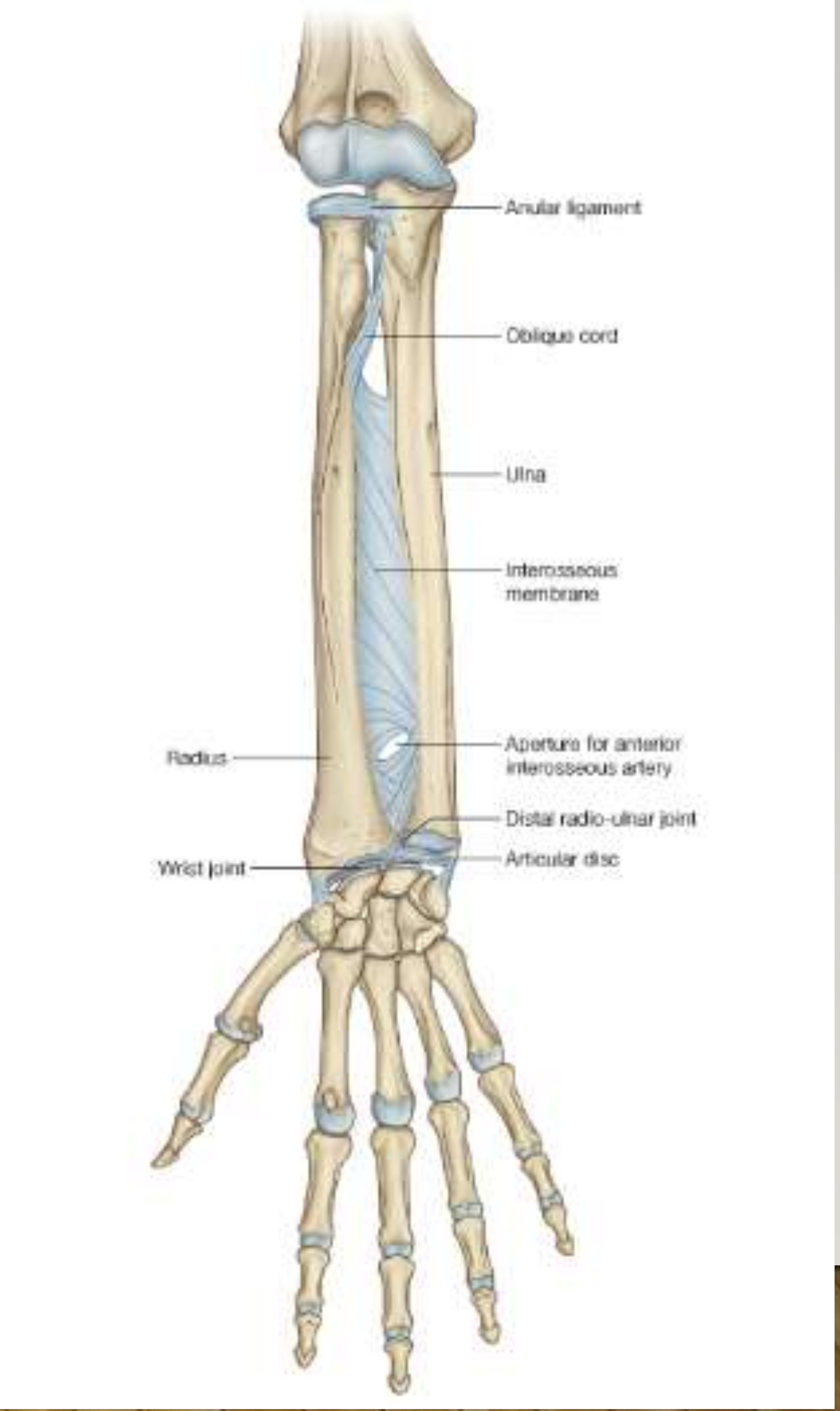



A
distal
middle
proximal
Phalanges
distal
proximal
Metacarpals
V
IV
III
II
I
Capitate
Hook of
hamate
Hamate
Carpal bones
Pisiform
Triquetrum
Lunate
Ulna
Trapezoid
Tubercle of trapezium
Trapezium
Tubercle of scaphoid
Scaphoid
Carpal bones
Wrist joint
Radius

THE ULNA

- It is the medial bone of the forearm.
- It is a **long bone**, with a shaft, an upper and a lower end.
- **Articulations:**

1. The radial notch of the upper end of the ulna articulates with the head of the radius to form the **superior radio-ulnar joint.**
2. The trochlear notch of the upper end of the ulna articulates with the trochlea of the lower end of the humerus (**elbow joint**).
3. The head of the lower end of the ulna articulates with the ulnar notch of the radius to form the **inferior radio-ulnar joint**.



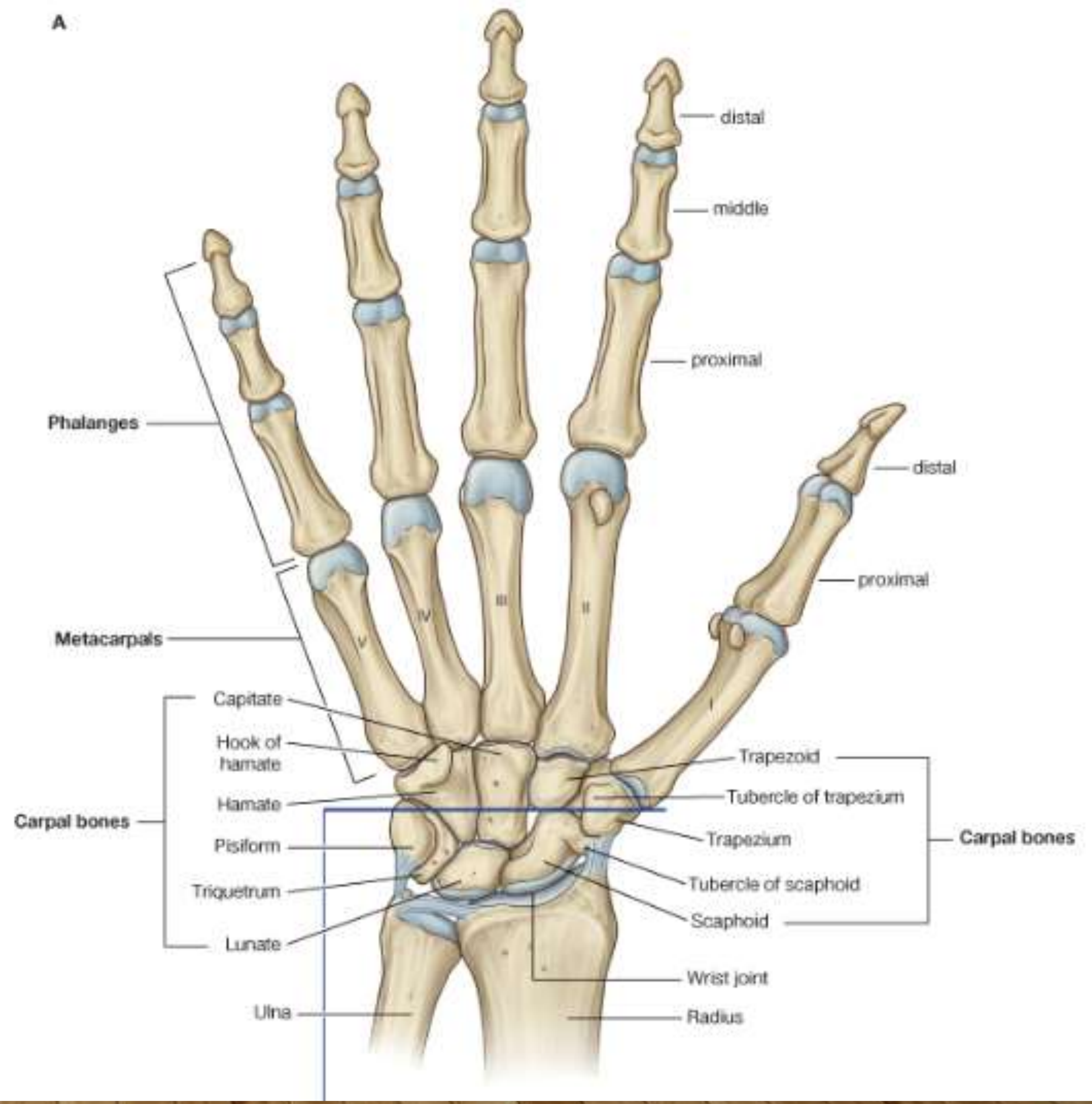

Trochlear notch
Olecranon
Coronoid process
Anterior view
Radial notch
Tuberosity of ulna
Anterior surface
Anterior surface
Anterior border (rounded)
Interosseous border
Medial surface
Posterior border (sharp)
Interosseous border
Posterior surface
Anterior border
Roughening for attachment of pronator quadratus
Ulnar styloid process
Attachment of articular disc
Distal view

SKELETON OF THE HAND

➢ The skeleton of the wrist and hand is composed of:

1. **Carpal bones.**
2. **Metacarpal bones.**
3. **Phalanges.**

➢ The carpal bones are **eight short bones** arranged in 2 rows: proximal and distal. The metacarpal bones are **five** in number. The phalanges are **fourteen** in number three for each finger and two for the thumb



…………….bone is one of bones that form the shoulder girdle

………….. Bone is the bone of the arm

…………….bone is the medial bone in the forearm

There are …….carpal bones in the each upper limb of human body

Head of humerus with glenoid cavity of scapula form …….joint
Radius meets with carpal bones forming…….. joint

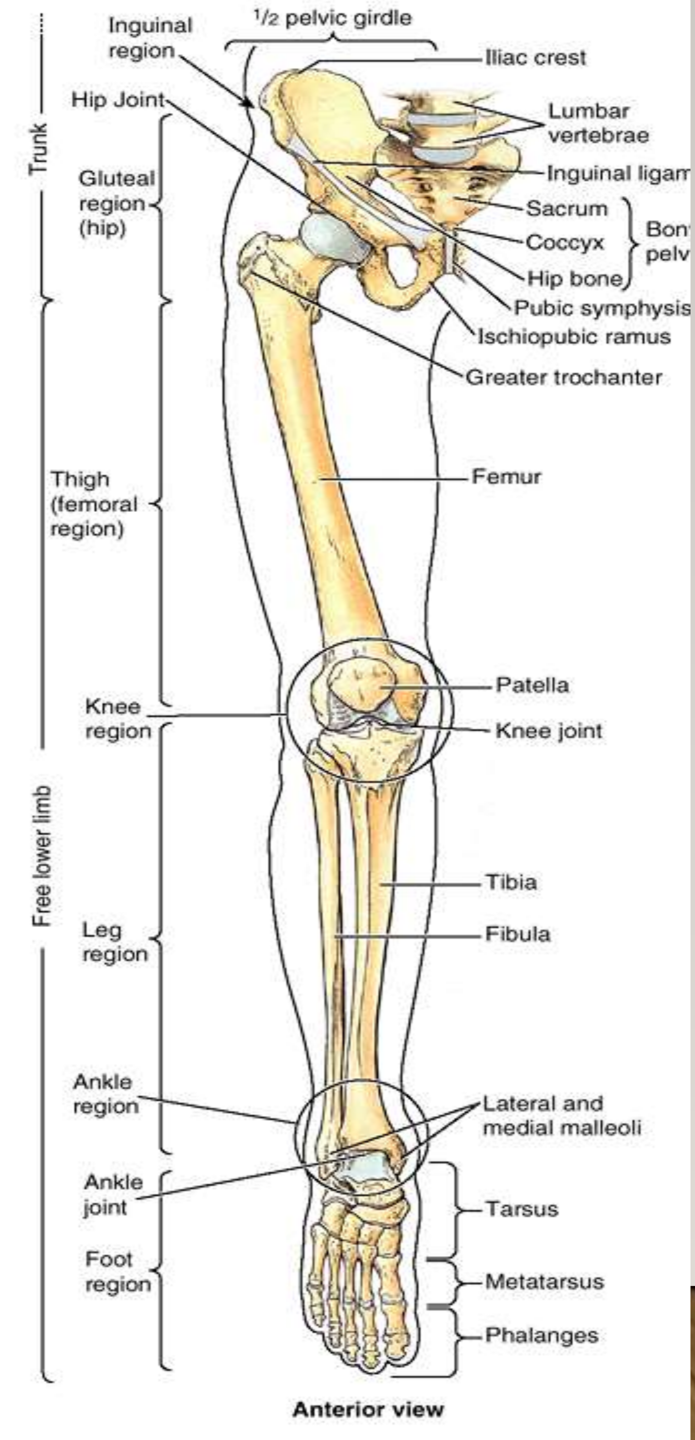
BONES OF LOWER LIMB

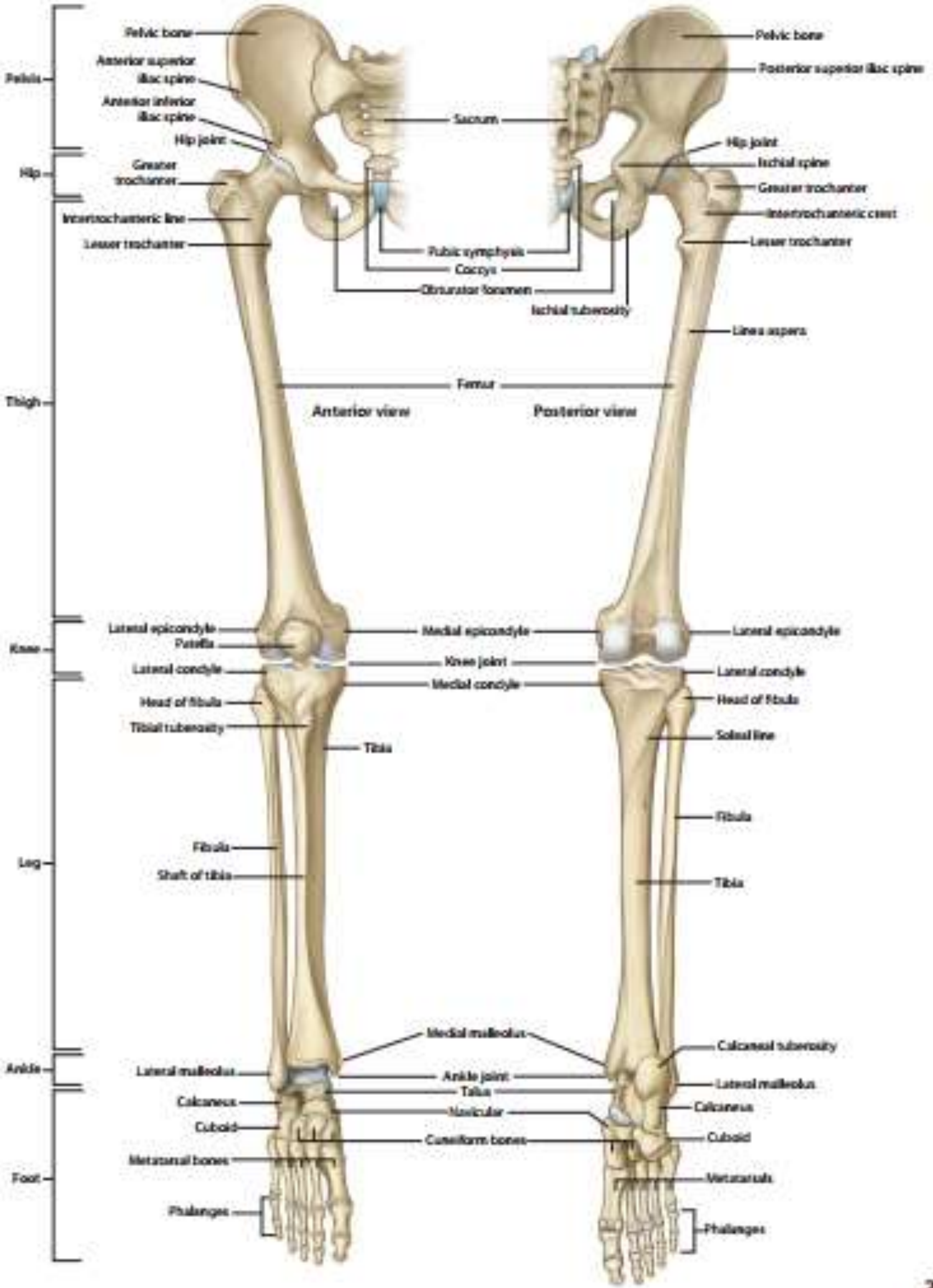
The lower limb is composed of the pelvic girdle, thigh, leg and foot.

The bones of these segments are as follows:

1. The bone of **the pelvic girdle**: is the **hip bone**.
2. The bone of the **thigh**: is the **femur**.
3. Bones of the **leg** are **tibia** (medially), and **fibula** (laterally).
4. Bones of the **foot: are the tarsus, metatarsus and phalanges.**

NB: Patella in the knee region.

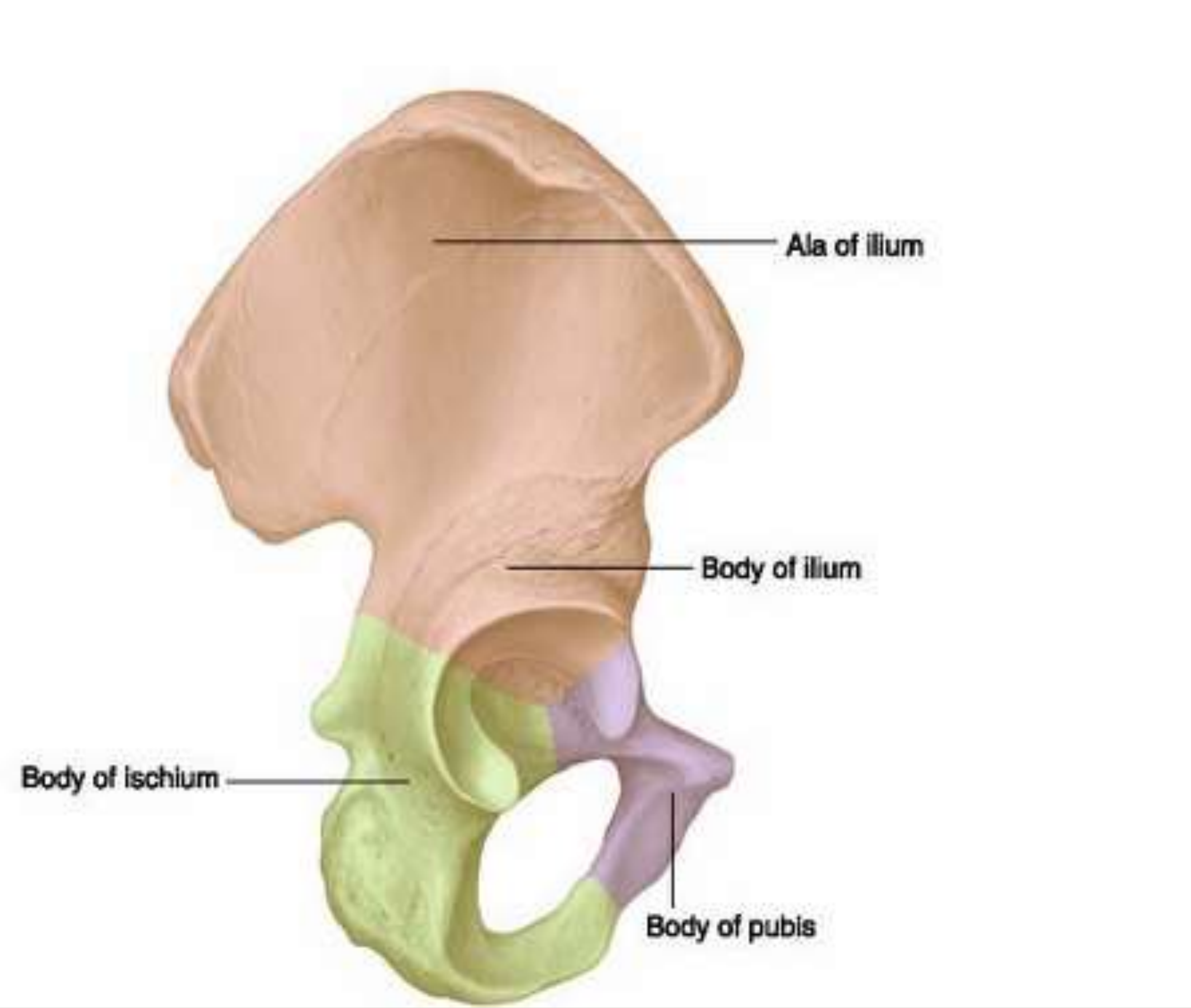


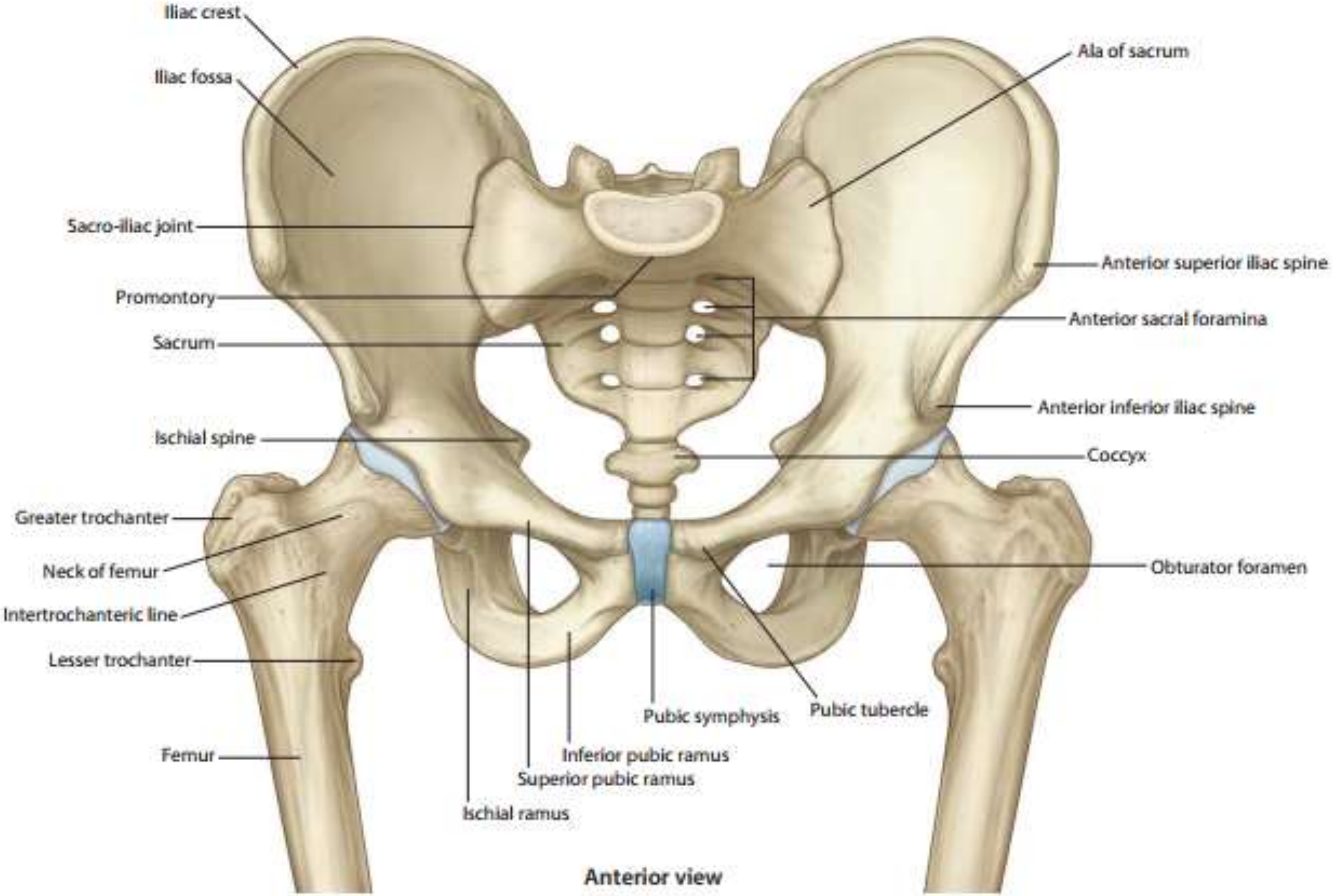

Pelvis
Hip
Thigh
Knee
Leg
Ankle
Foot
Pelvic bone
Anterior superior iliac spine
Anterior inferior iliac spine
Hip joint
Greater trochanter
Intertrochanteric line
Lesser trochanter
Sacrum
Pubic symphysis
Coccyx
Obturator foramen
Ischial tuberosity
Femur
Anterior view
Posterior view
Pelvic bone
Posterior superior iliac spine
Hip joint
Ischial spine
Greater trochanter
Intertrochanteric crest
Lesser trochanter
Linea aspera
Lateral epicondyle
Patella
Lateral condyle
Head of fibula
Tibial tuberosity
Medial epicondyle
Knee joint
Medial condyle
Tibia
Lateral epicondyle
Lateral condyle
Head of fibula
Soleal line
Fibula
Fibula
Shaft of tibia
Tibia
Medial malleolus
Lateral malleolus
Ankle joint
Talus
Calcaneus
Navicular
Cuboid
Cuneiform bones
Metatarsal bones
Phalanges
Calcaneal tuberosity
Lateral malleolus
Calcaneus
Cuboid
Metatarsals
Phalanges

HIP BONE

- It is a large, **irregular bone**.
- It is formed of three parts, ileum, ischium and pubis fused together in the acetabulum which is a cup- shaped hollow on the lateral surface of the bone.
- Below and anterior to it is the obturator foramen.
- **Articulations:**

1. By the acetabulum with the head of femur (**hip joint**).
2. By the pubis with the pubis of opposite side (**symphysis pubis**), a secondary cartilagenous joint.
3. By the posterior part of ileum with the sacrum (**sacroiliac joint**).



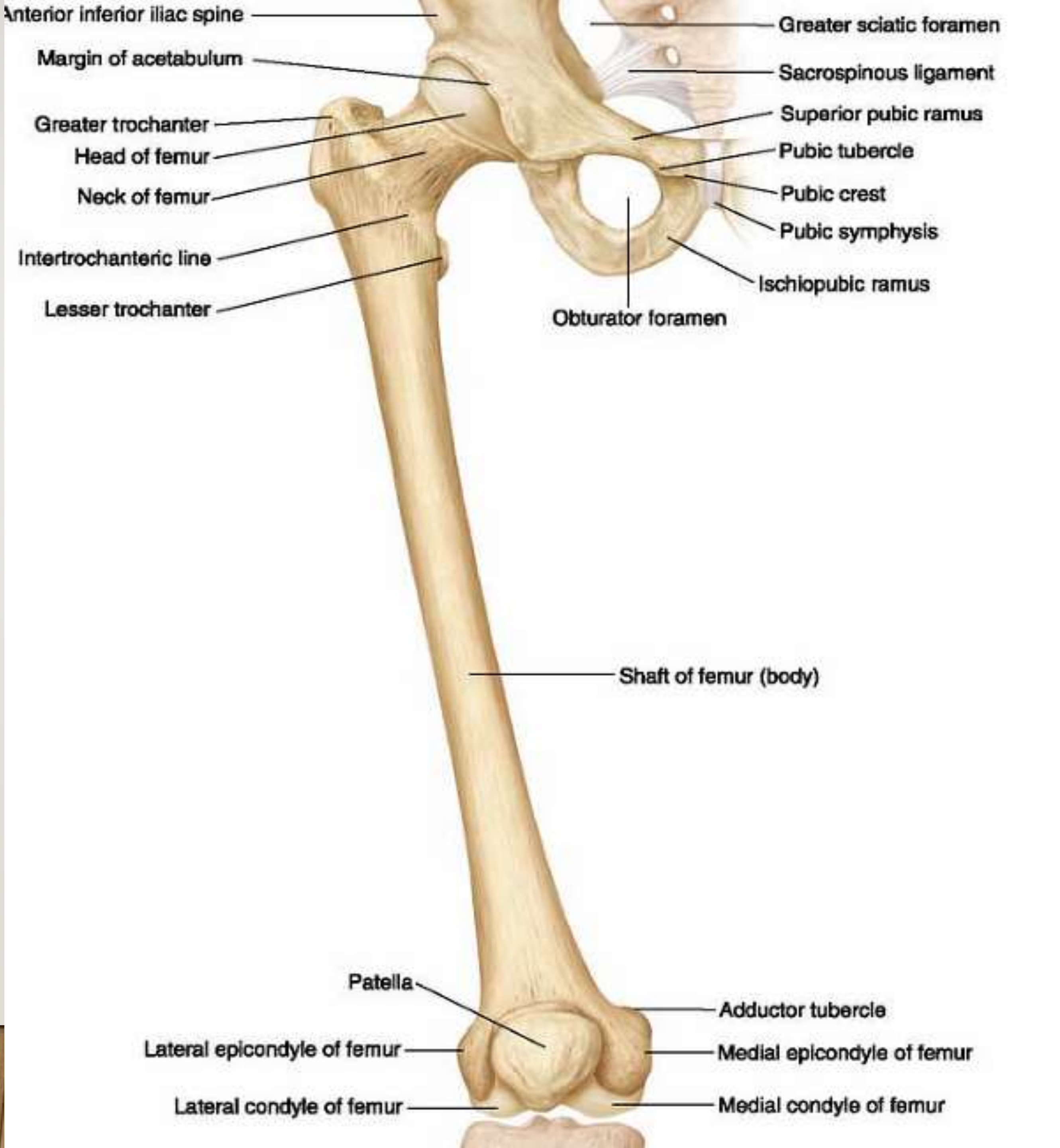


Anterior view

THE FEMUR

- It is the largest and strongest bone in the body.
- It is a **long bone**; consists of a shaft, upper and lower ends.
- **Articulations:**

1. The head articulates with the acetabulum of the hip bone to form the **hip joint**.
2. The two condyles of the lower end articulate with the tibia in the **knee joint**.
3. The patella articulates with the anterior surface of the lower end of femur **(knee joint)**.

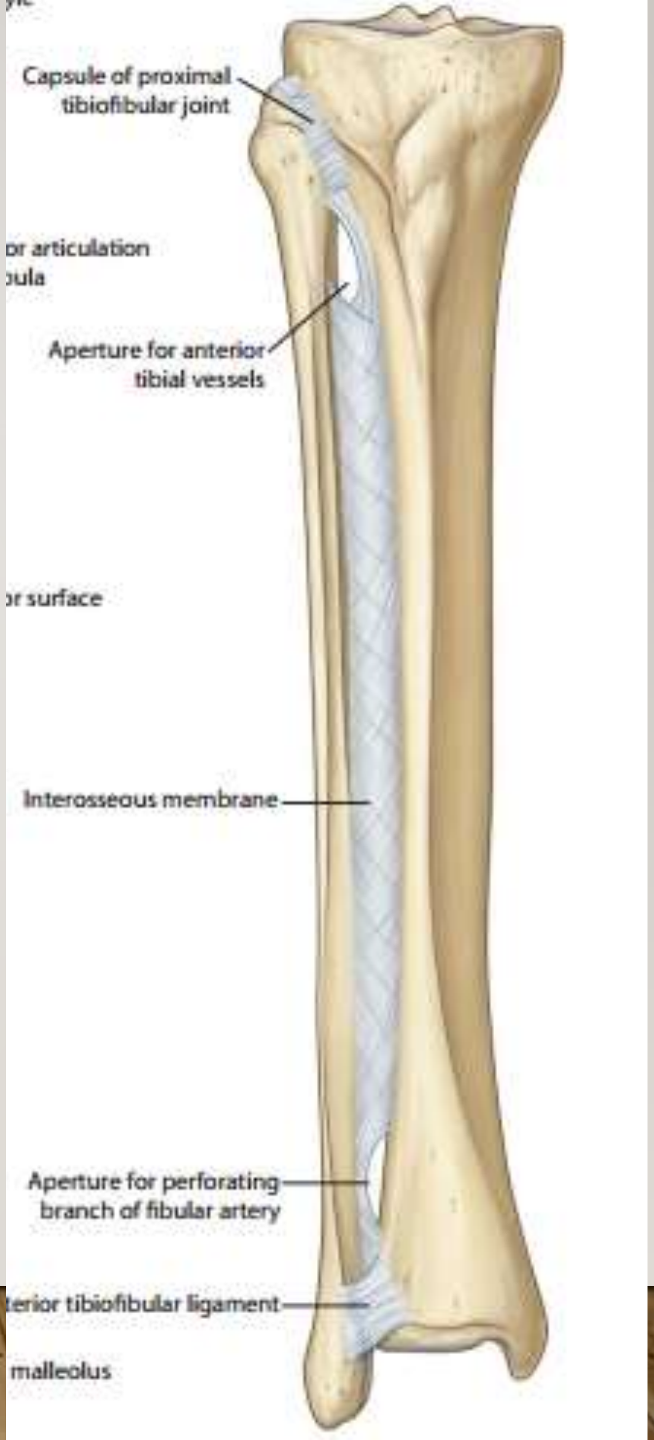


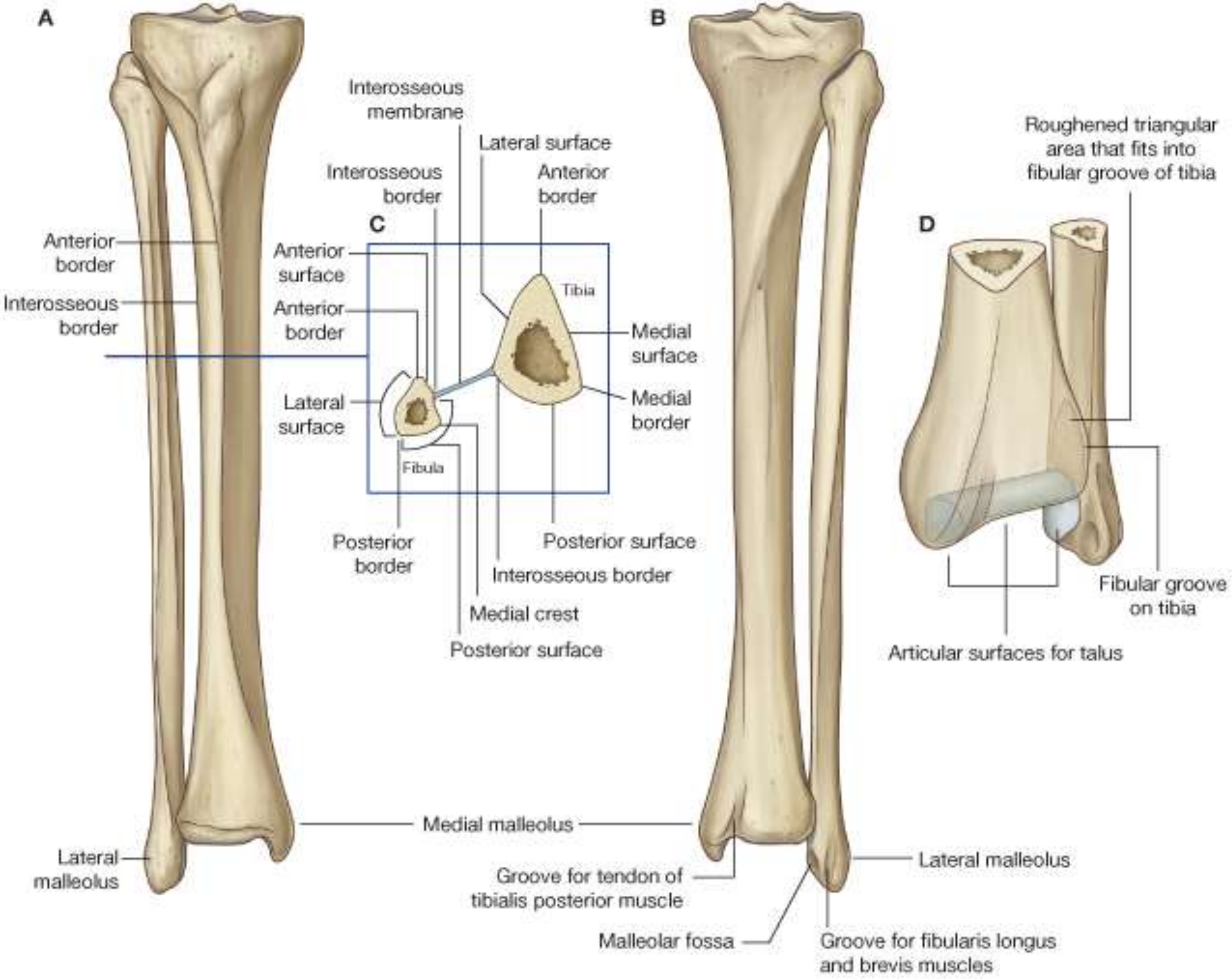

KNEE BONES
Femur
Patella
Tibia
VERITAS health

THE TIBIA

- It is the medial bone of the leg.
- It is a **long bone**; made up of a shaft, an upper and lower ends.
- **Articulations:**

1. By the superior surface of medial and lateral condyles of the tibia with the condyles of femur (**knee joint**).
2. By the inferior surface of lower end of the tibia with the talus (**ankle joint**).
3. By the lateral side of lateral condyle of the tibia with the head of fibula to form **superior tibiofibular joint**.
4. By lateral side of lower end of the tibia with the lowermost part of the shaft of fibula to form **inferior tibiofibular joint**.

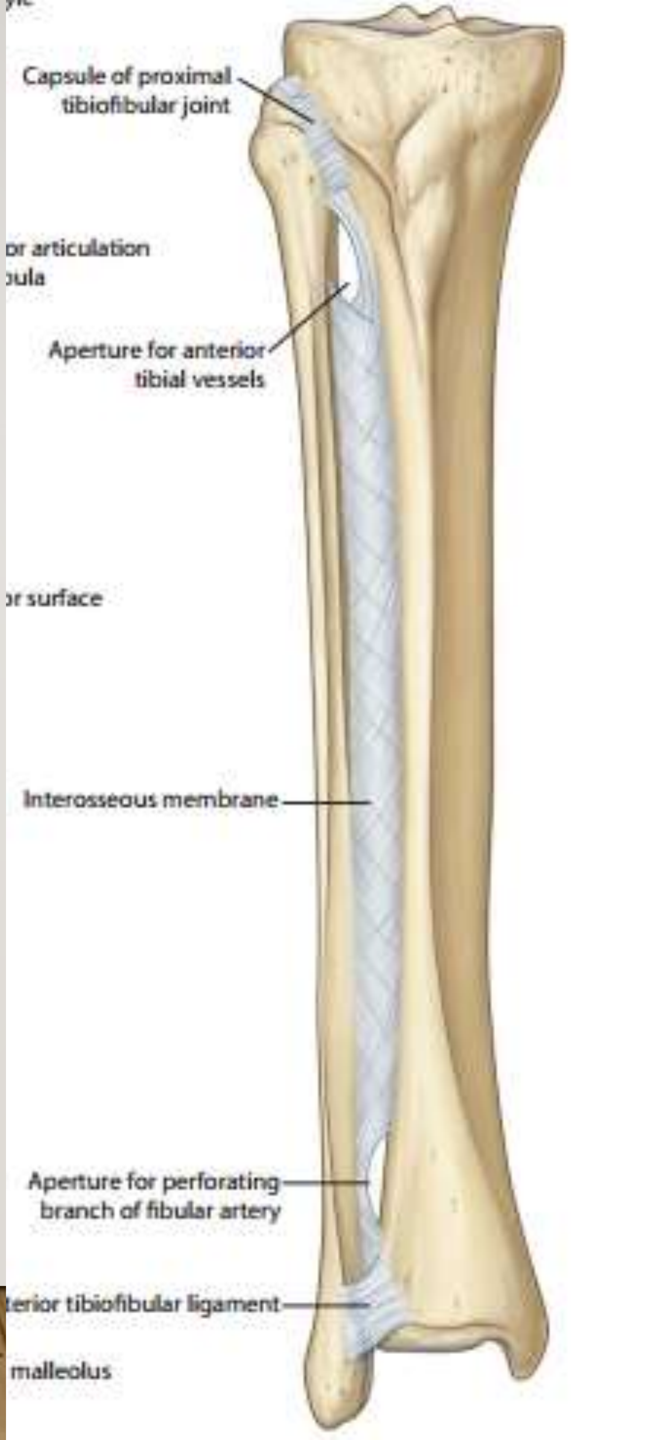


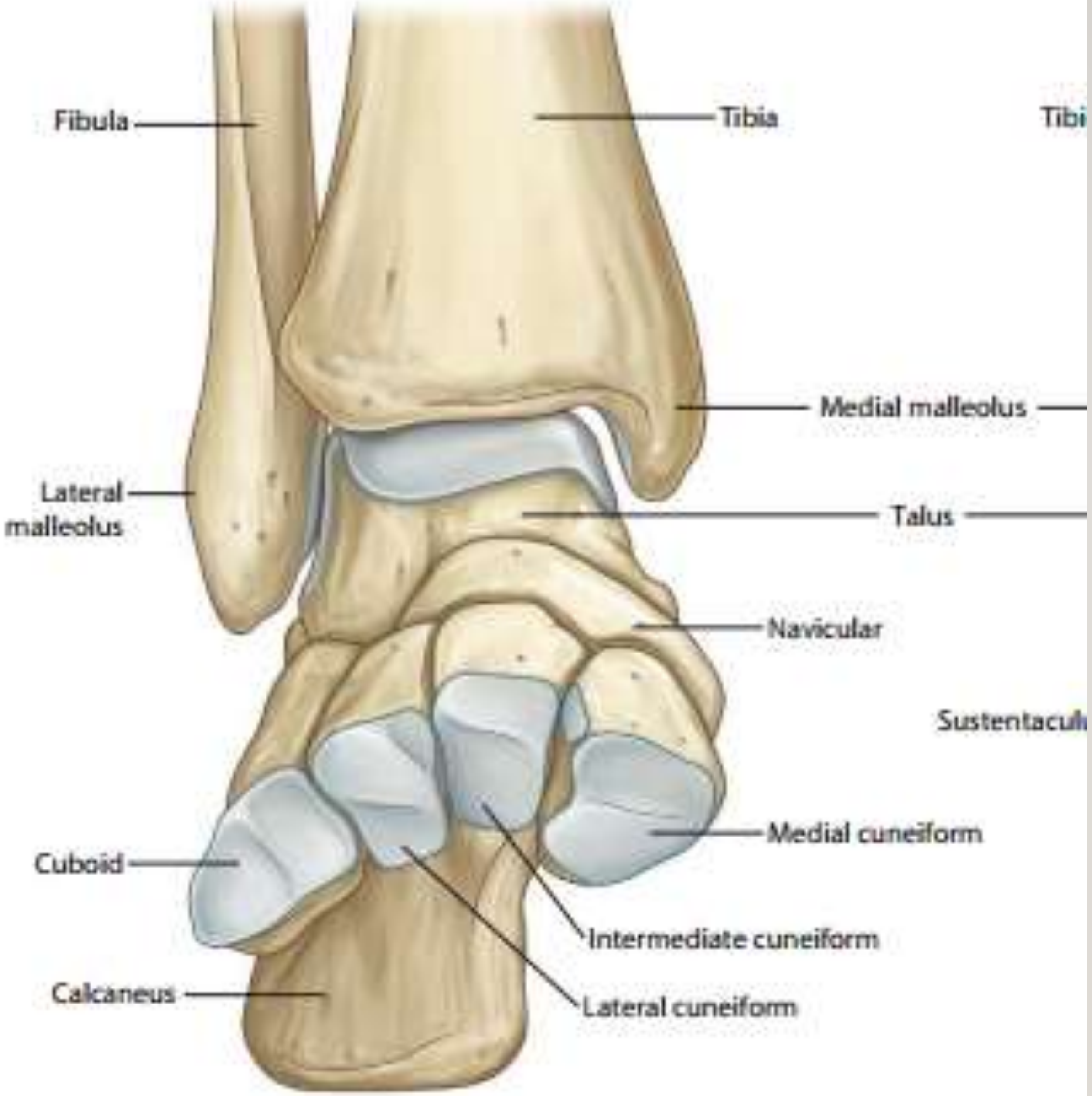

A
B
C
D
Interosseous membrane
Lateral surface
Interosseous border
Anterior border
Tibia
Anterior surface
Anterior border
Anterior border
Interosseous border
Medial surface
Medial border
Lateral surface
Fibula
Posterior border
Posterior surface
Interosseous border
Medial crest
Posterior surface
Roughened triangular area that fits into fibular groove of tibia
Fibular groove on tibia
Articular surfaces for talus
Medial malleolus
Lateral malleolus
Lateral malleolus
Groove for tendon of tibialis posterior muscle
Malleolar fossa
Groove for fibularis longus and brevis muscles

THE FIBULA

- It is the lateral bone of the leg.
- It is a **long bone**; made up of upper end, a shaft, and lower end (lateral malleolus).
- **Articulations:**

1. The head of the fibula articulates with the tibia to form **superior tibiofibular joint**.
2. By its lowermost medial part of the shaft, it articulates with the tibia to form **inferior tibiofibular joint**.
3. By the articular part of lateral malleolus, it articulates with talus at the **ankle joint**.



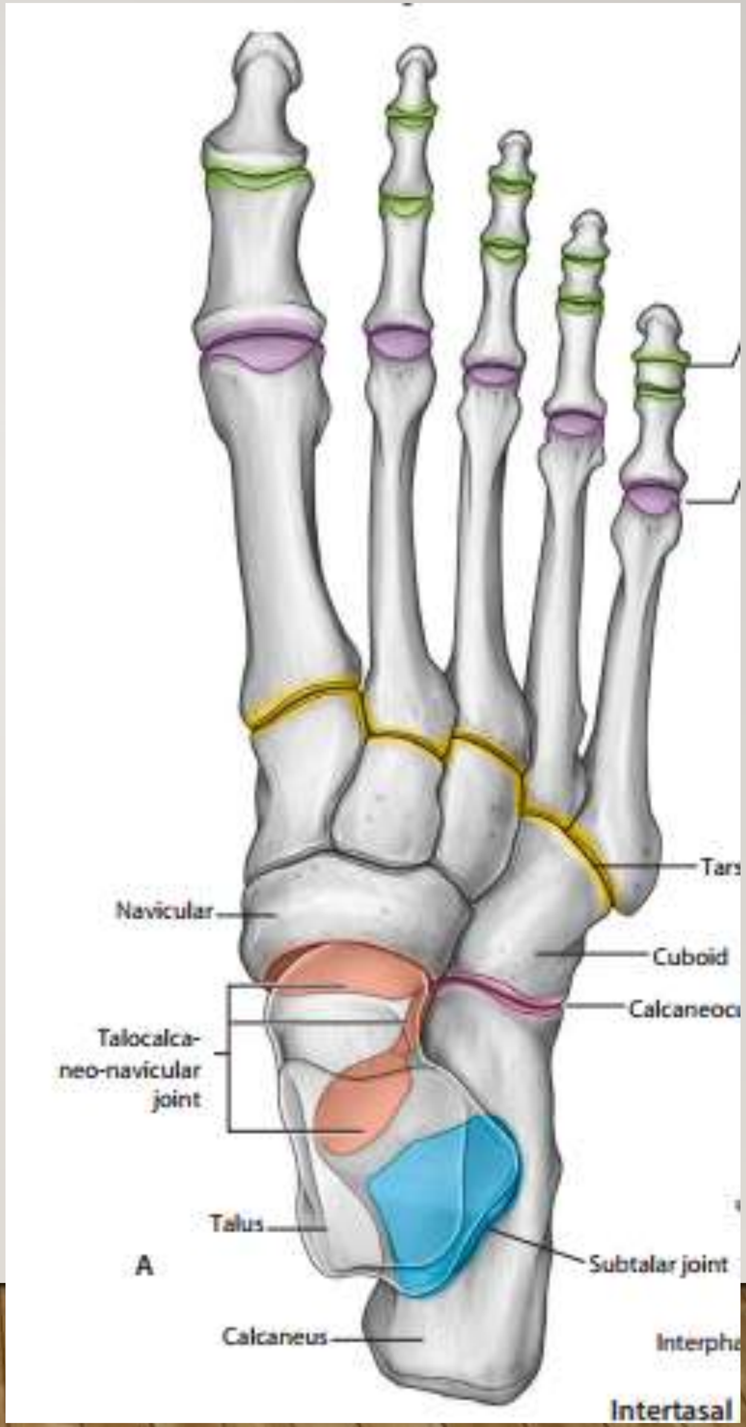


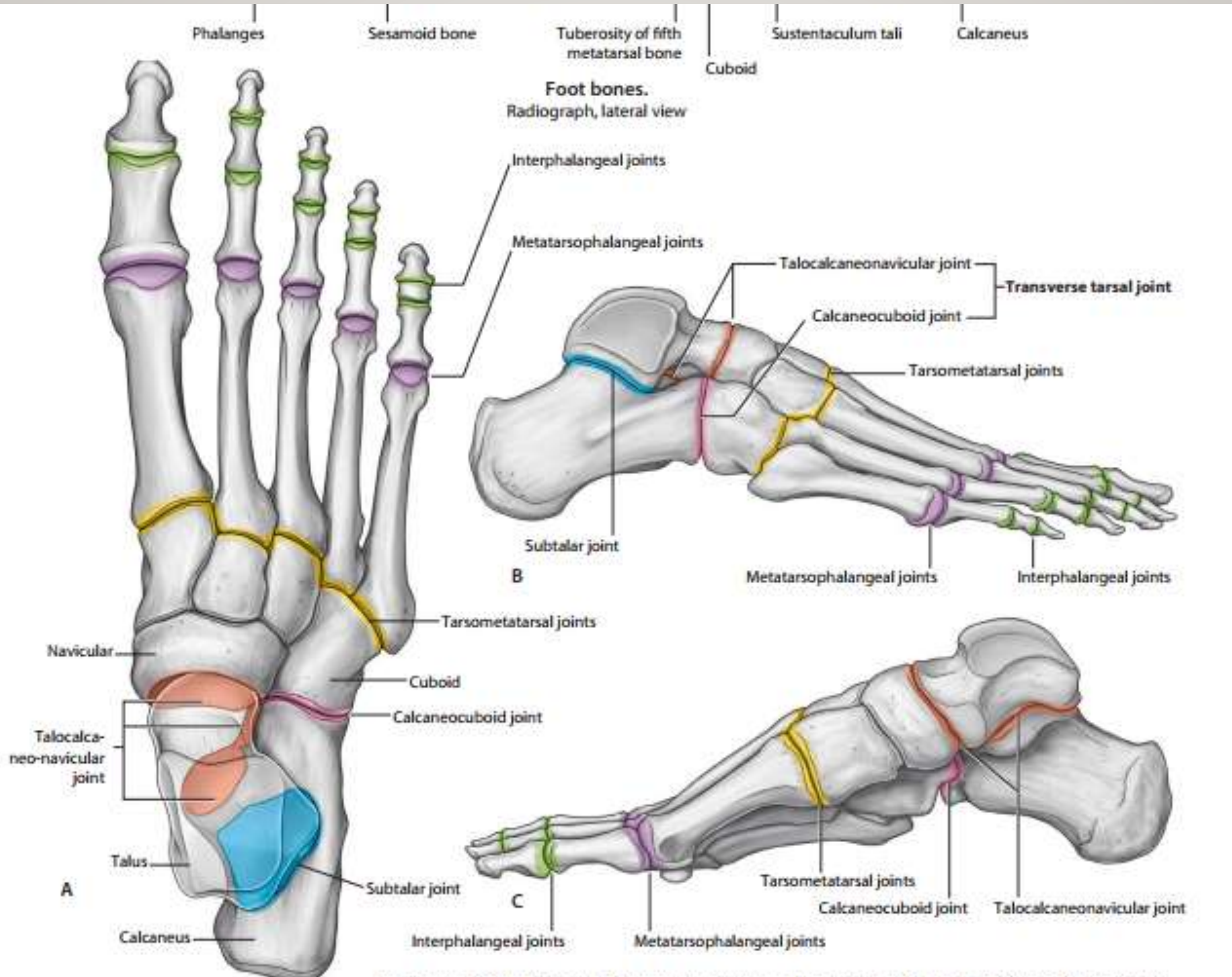
Anterior view
(metatarsals and phalanges removed)

SKELETON OF THE FOOT

- The bones of the foot are made of:

1. The **tarsus** bones (**7 short bones**) form the posterior part of the foot.
2. The **metatarsus** bones (5 bones) form the anterior part of the foot.
3. The **phalanges** bones of toes (14 bones); two in the big toe and three in each of the other toes.





Intertasal joints: A. Dorsal view (talus drawn as transparent) B. Lateral view C. Medial view

Number of metatarsals is……….in one limb

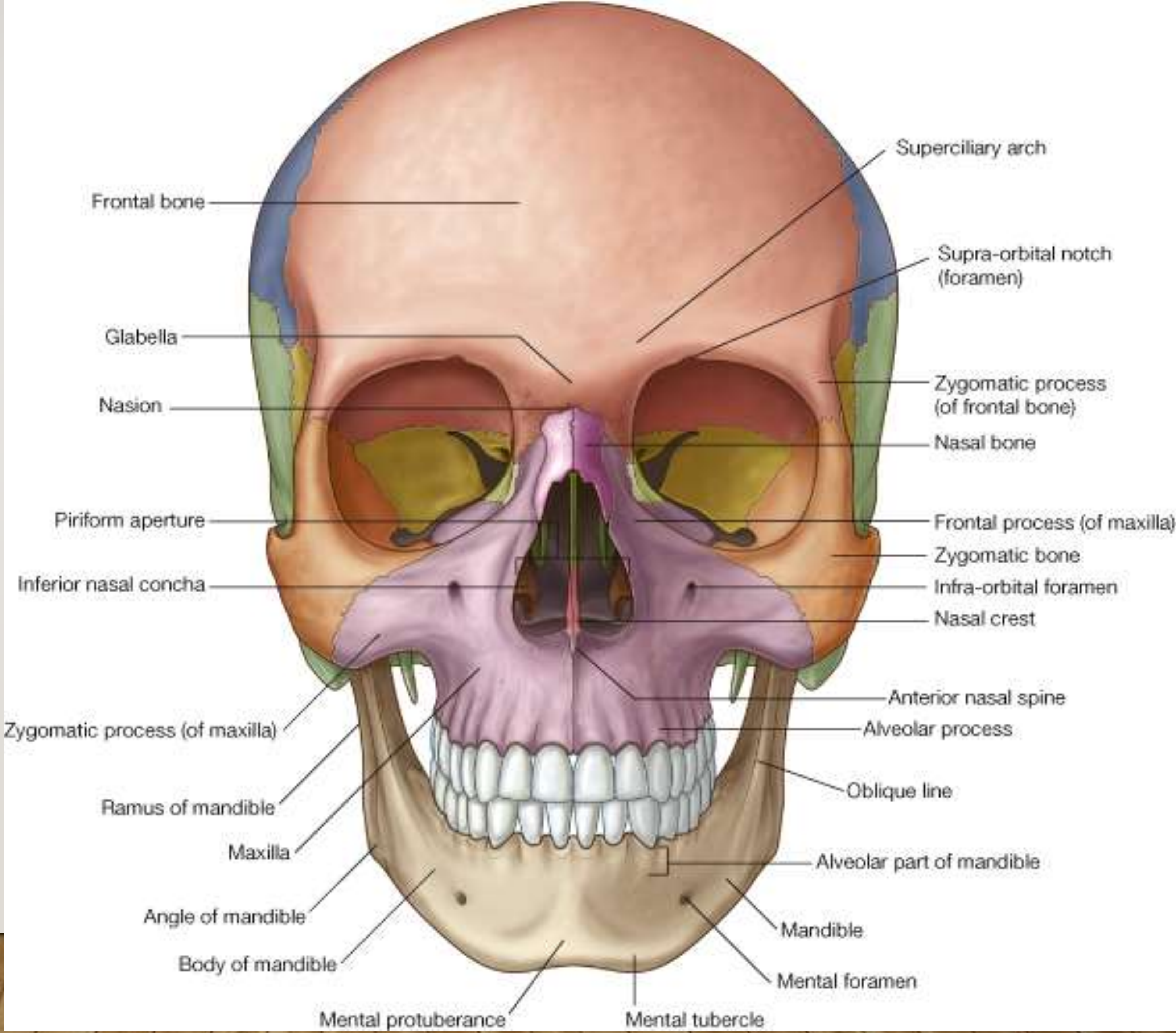
Head of the femur articulates with acetabulum of hip bone forming …… joint

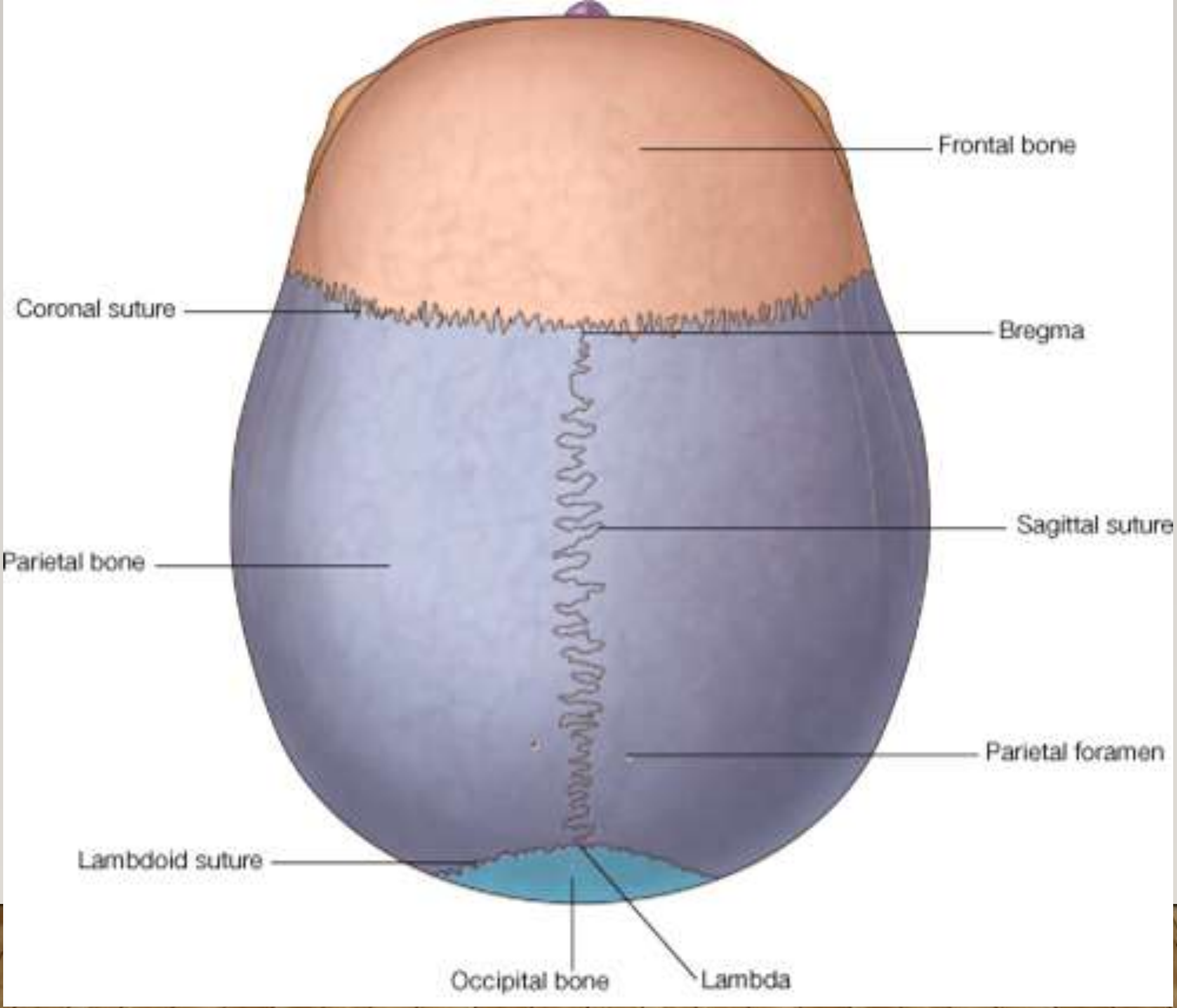
Condyles of the femur articulate with condyles of tibia forming……. Joint

THE SKULL

- It is the skeleton of the head.
- It is formed of the mandible and other some immovable flat bones articulate with each other by fibrous joint.
- Some important features:

Frontal bone
Parietal bone
Occipital bone
Maxilla
Mandible
Orbit
the anterior aperture of the nasal cavity
The cranial cavity: It is the interior of the skull.
Sutures

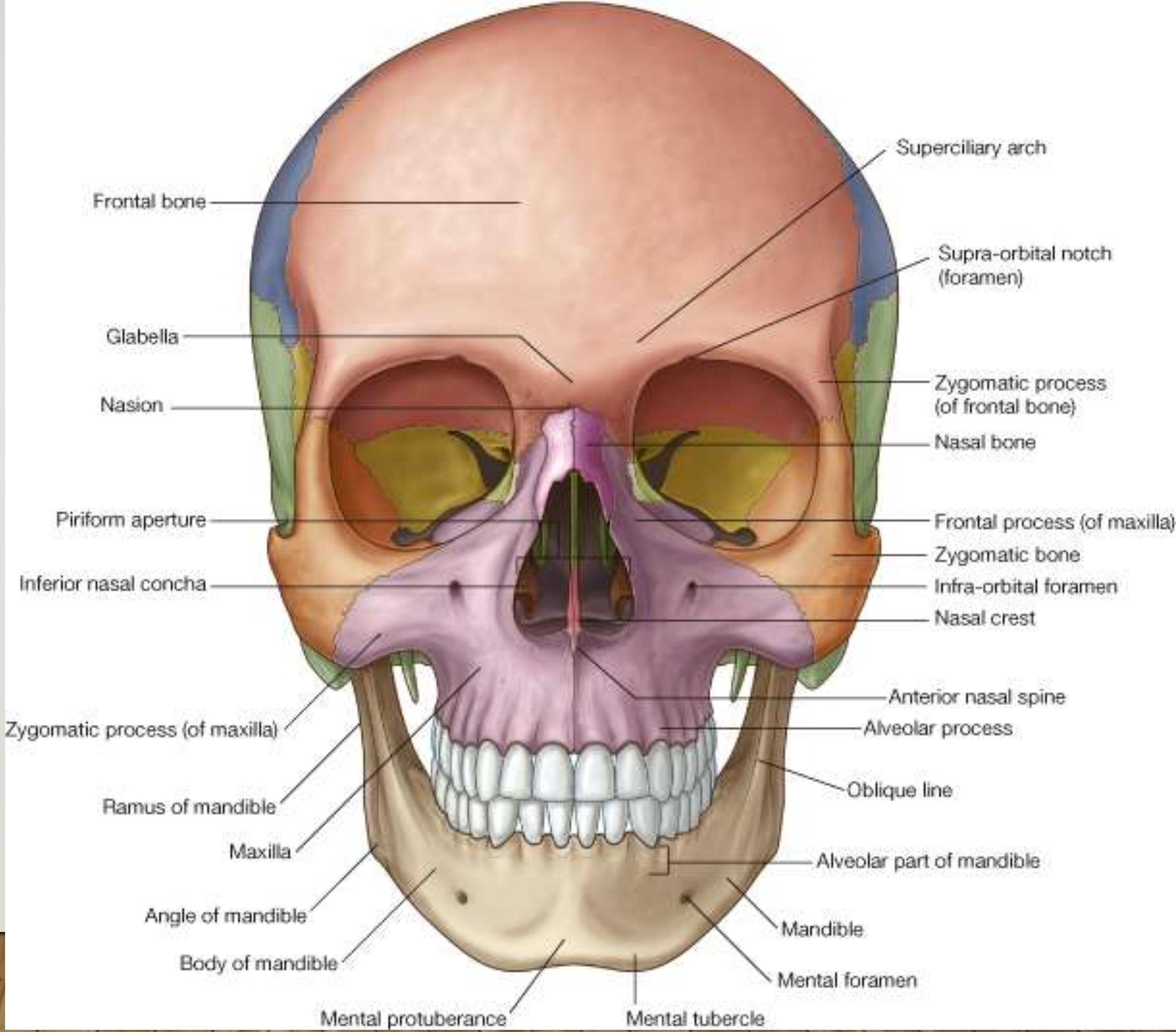


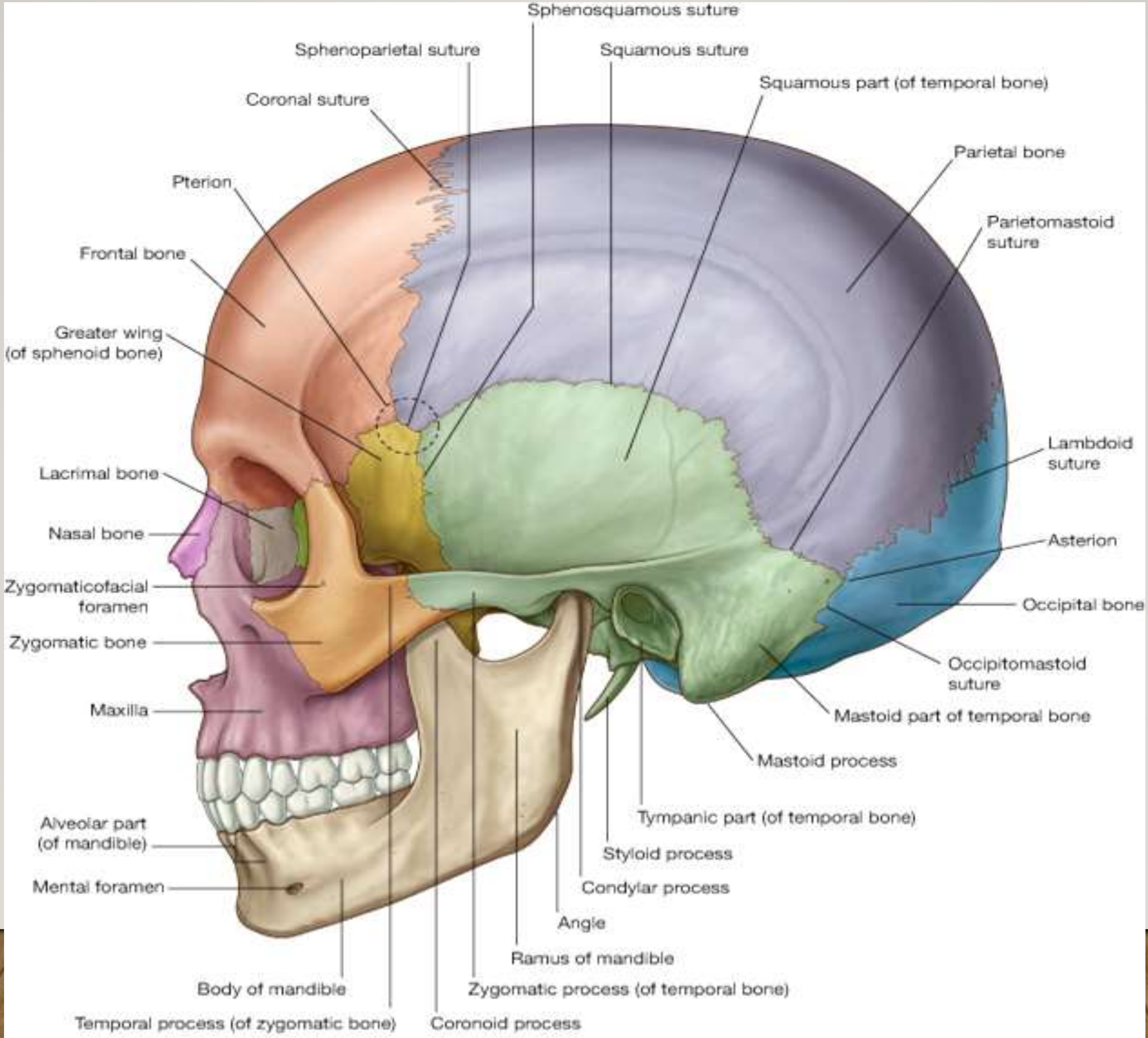


Frontal bone
Coronal suture
Bregma
Sagittal suture
Parietal bone
Parietal foramen
Lambdoid suture
Occipital bone
Lambda

THE MANDIBLE

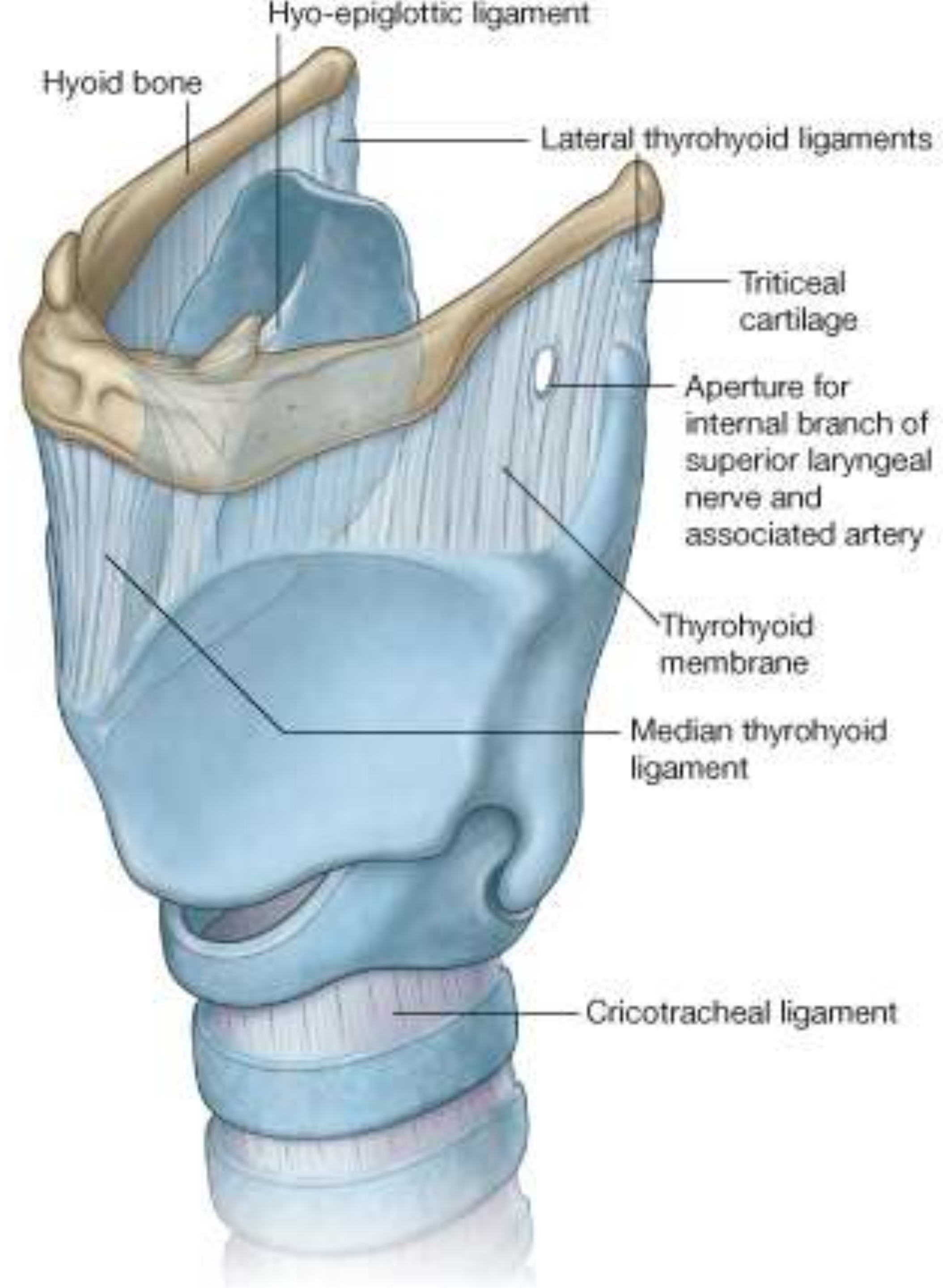
- It is freely movable bone. It is formed of a body and two rami.
- The upper surface of the mandible lodges the teeth.
- It articulates with the skull at **the temporomandibular joints**.




Sphenosquamous suture
Sphenoparietal suture
Squamous suture
Squamous part (of temporal bone)
Coronal suture
Parietal bone
Pterion
Parietomastoid suture
Frontal bone
Greater wing (of sphenoid bone)
Lambdoid suture
Lacrimal bone
Nasal bone
Asterion
Zygomaticofacial foramen
Occipital bone
Zygomatic bone
Occipitomastoid suture
Maxilla
Mastoid part of temporal bone
Mastoid process
Tympanic part (of temporal bone)
Alveolar part (of mandible)
Styloid process
Mental foramen
Condylar process
Angle
Ramus of mandible
Body of mandible
Zygomatic process (of temporal bone)
Temporal process (of zygomatic bone)
Coronoid process

THE HYOID BONE

- It is in the middle of the upper part of the front of the neck above the thyroid cartilage.



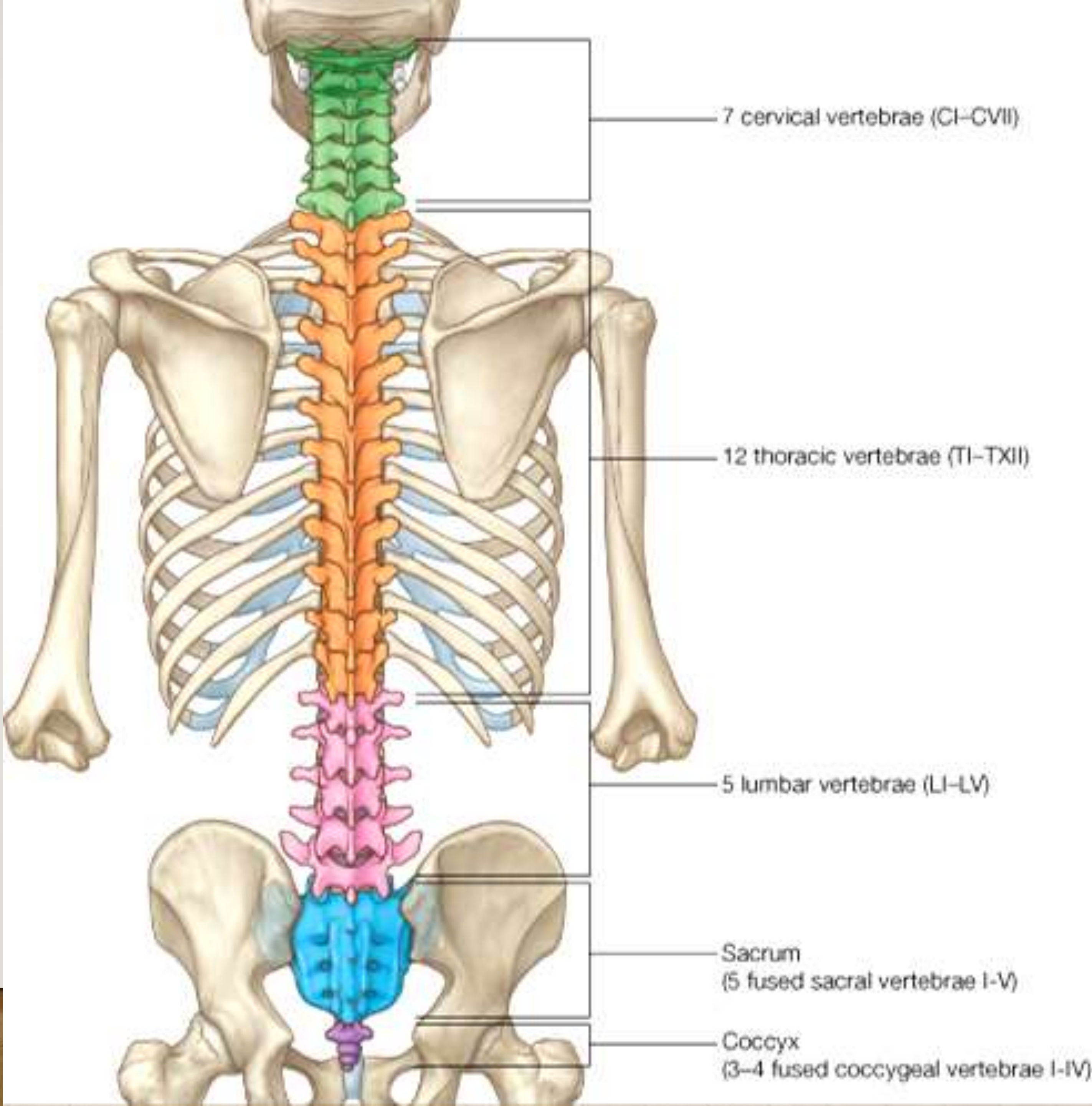
The human vertebral column

- It is made up of series of vertebrae: **7 cervical**, **12 thoracic**, **5 lumbar, 5 sacral** fused into the sacrum and **4 coccygeal** fused into the coccyx.

- Intervertebral disc lies inbetween the bodies of vertebrae.

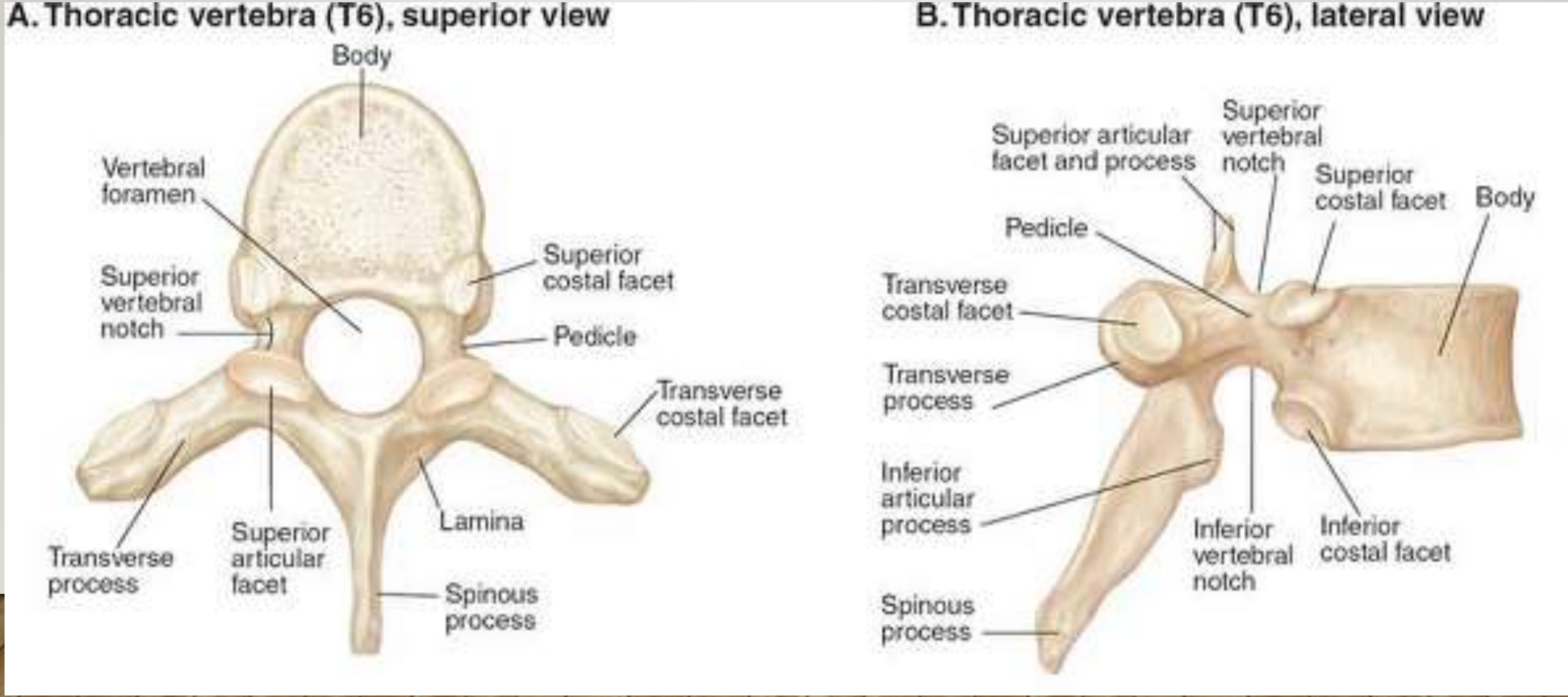
- Functions of the column:

1. Protects the spinal cord.
2. Supports the skull.
3. Transmits the weight of the body to the lower limbs.
4. Forms the axis of the body.



The human vertebral column

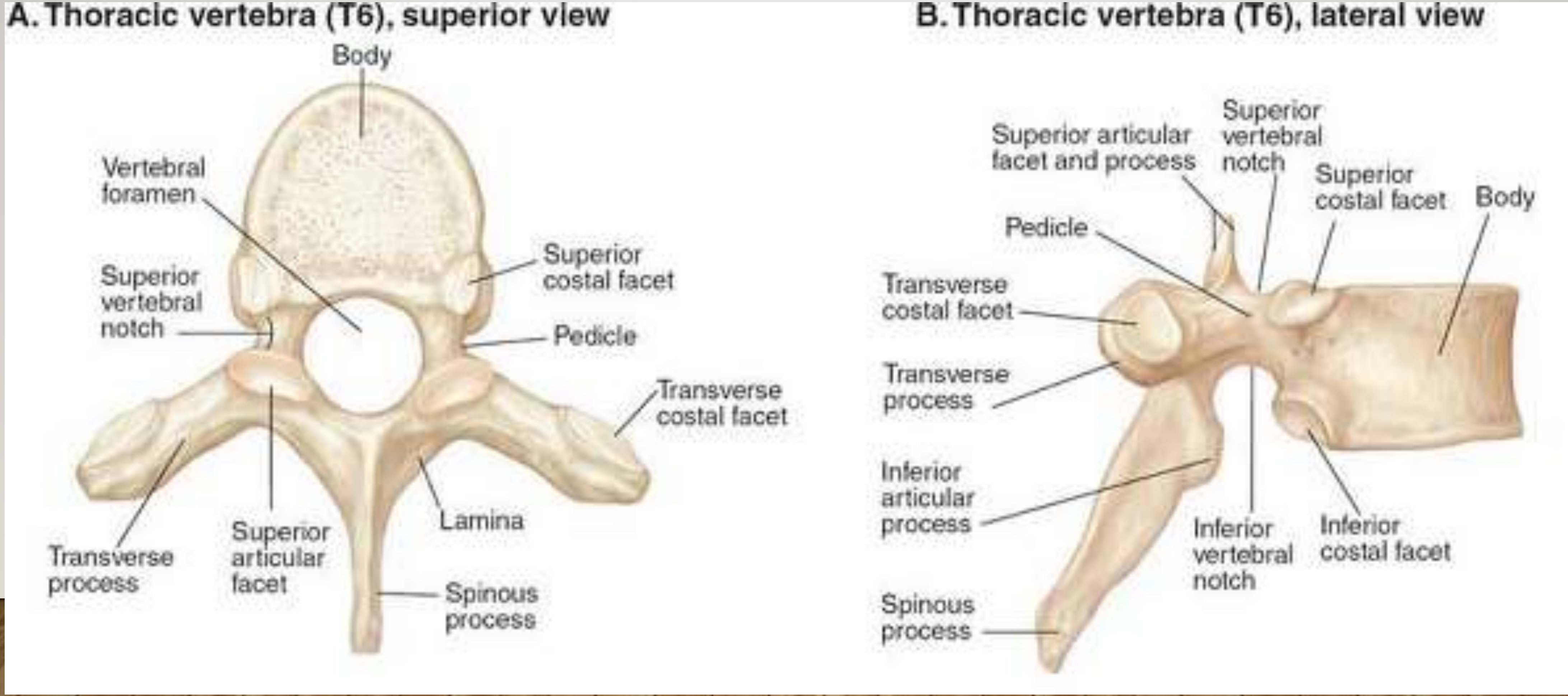
- Structure of the Typical Vertebra
- **Body**: Ventrally
- **Vertebral arch**: is made up of a pedicle and a lamina on each side.
- Between the posterior surface of the body and the neural arch is enclosed the **vertebral foramen** which contains the spinal cord and meninges.



The human vertebral column

- **Seven processes** diverge from the arch:

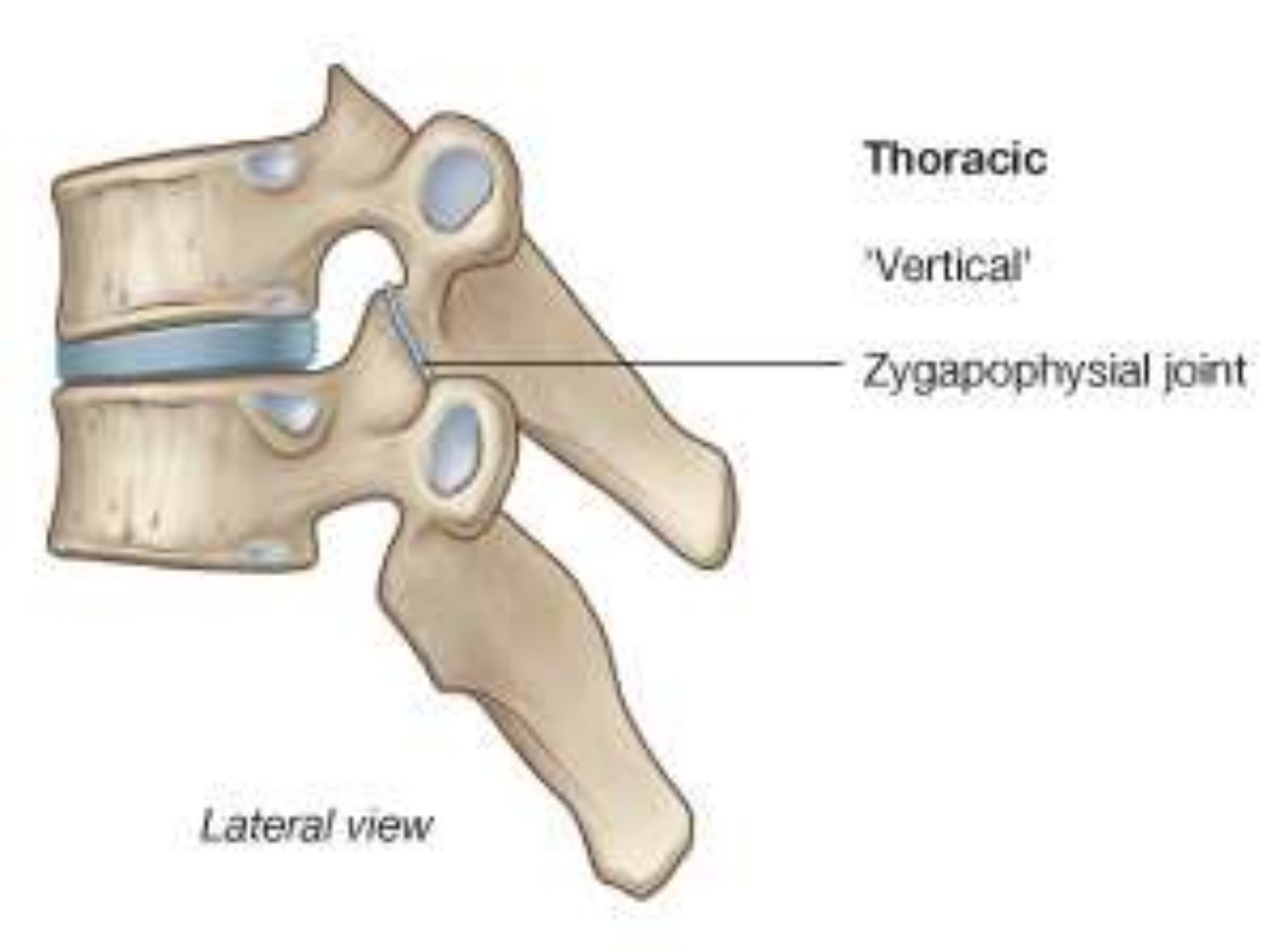
1. Spinous process, one in midline posteriorly.
2. Transverse processes, two projecting laterally on either sides.
3. Articular processes, two superior and two inferior.



The human vertebral column

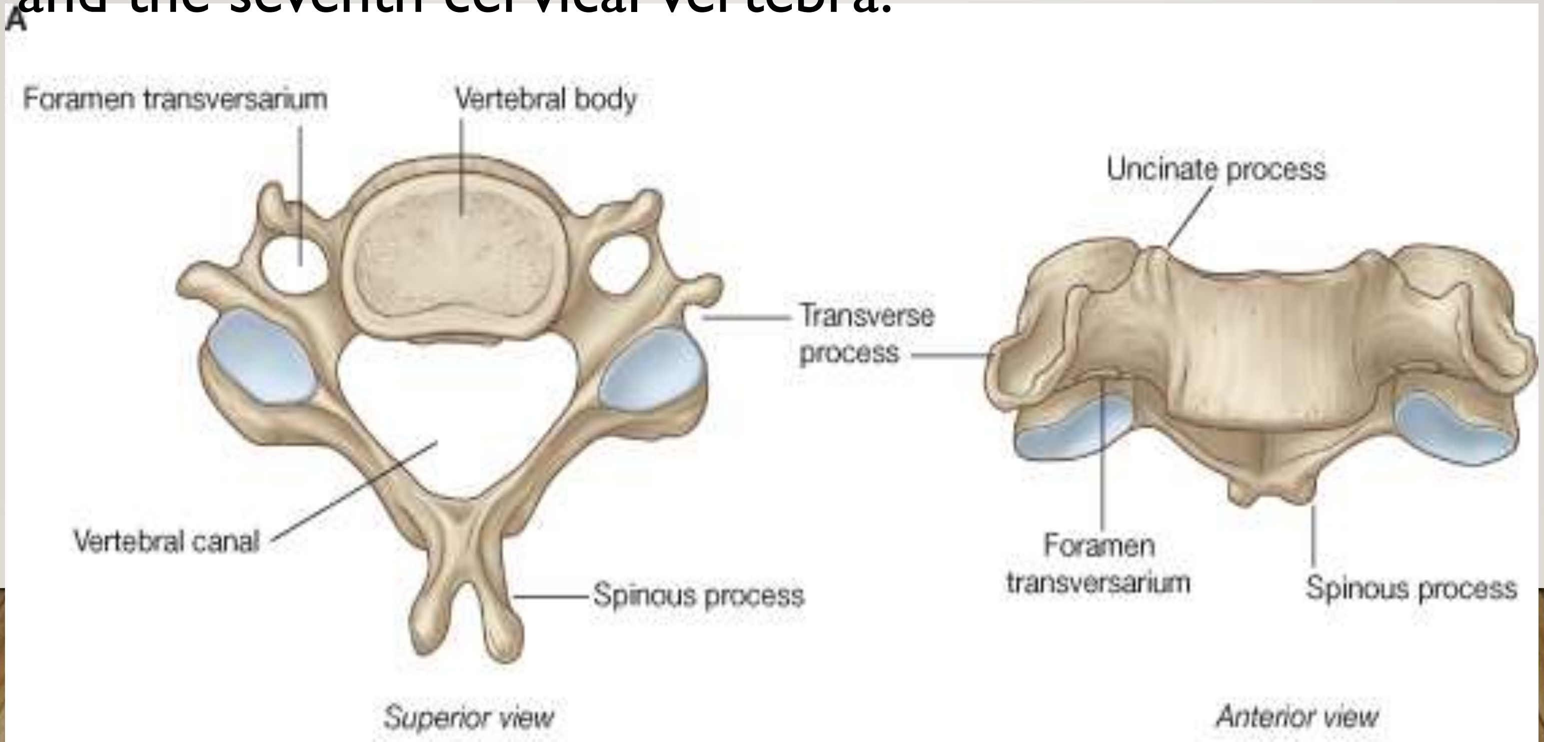
- **Articulations between two adjacent vertebrae:**

1. Bodies: articulate with each other by the intervertebral disc (secondary cartilaginous joint).
2. Articular processes articulate by synovial joints.

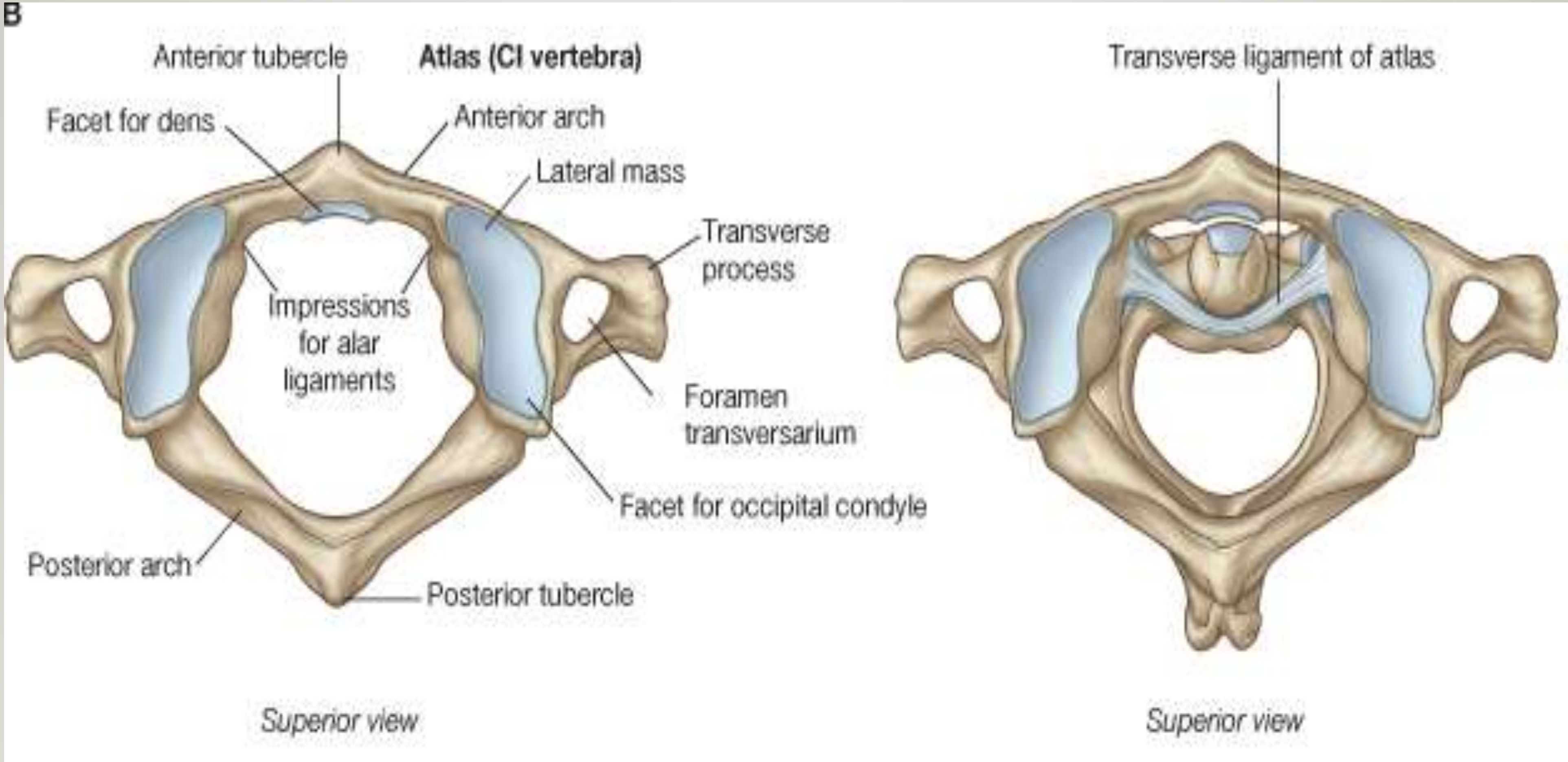


Cervical vertebrae:

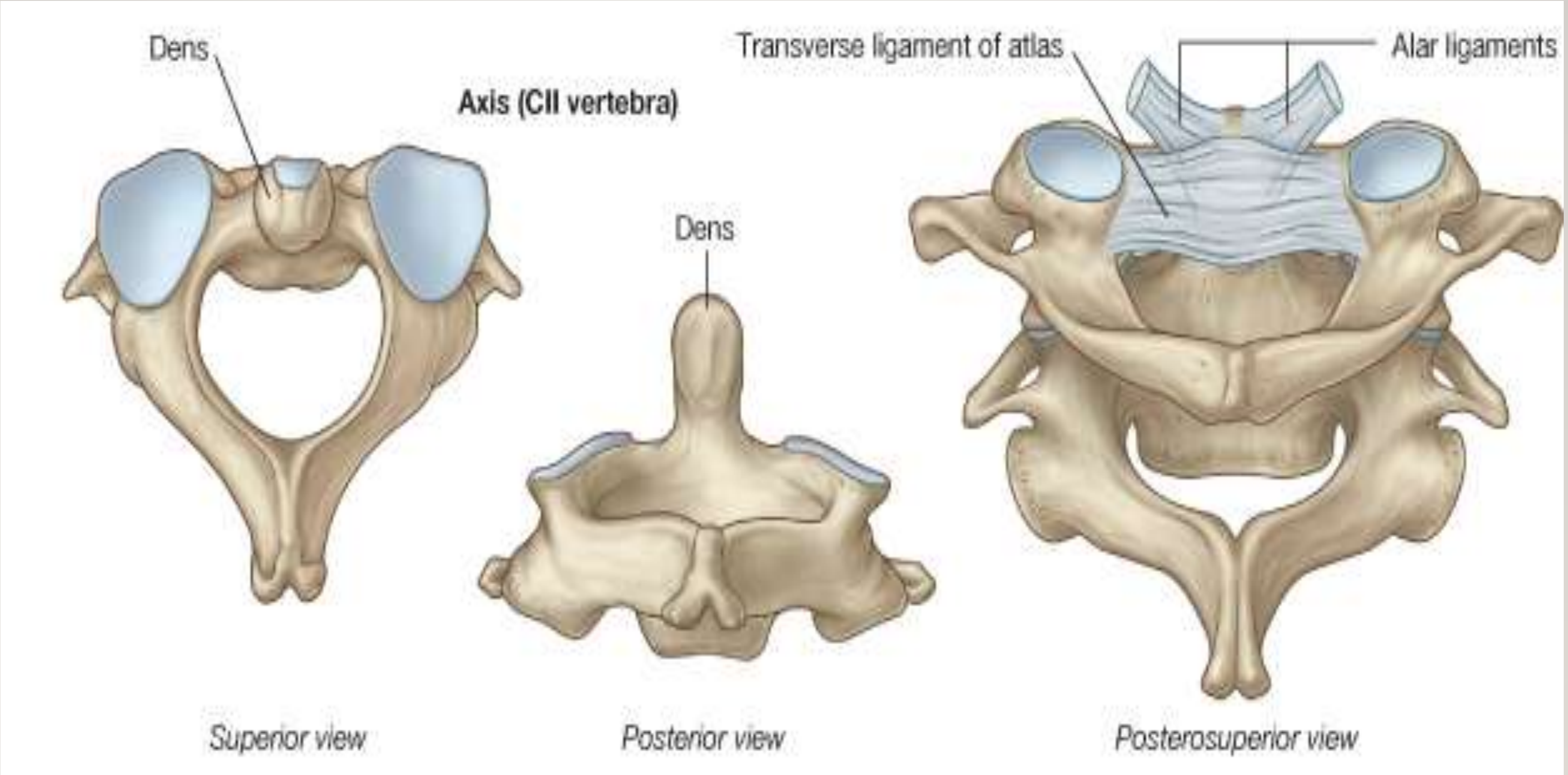
- From the 3rd to the 6th are typical.
- The first, second and seventh vertebra are atypical.
- Characters of typical cervical vertebra:
- The spine is short and bifid.
- The transverse processes have the foramen transversarium in which pass the vertebral vessels.
- The atypical cervical vertebrae, the first cervical vertebra (atlas), the second cervical vertebra (axis) and the seventh cervical vertebra.



atlas

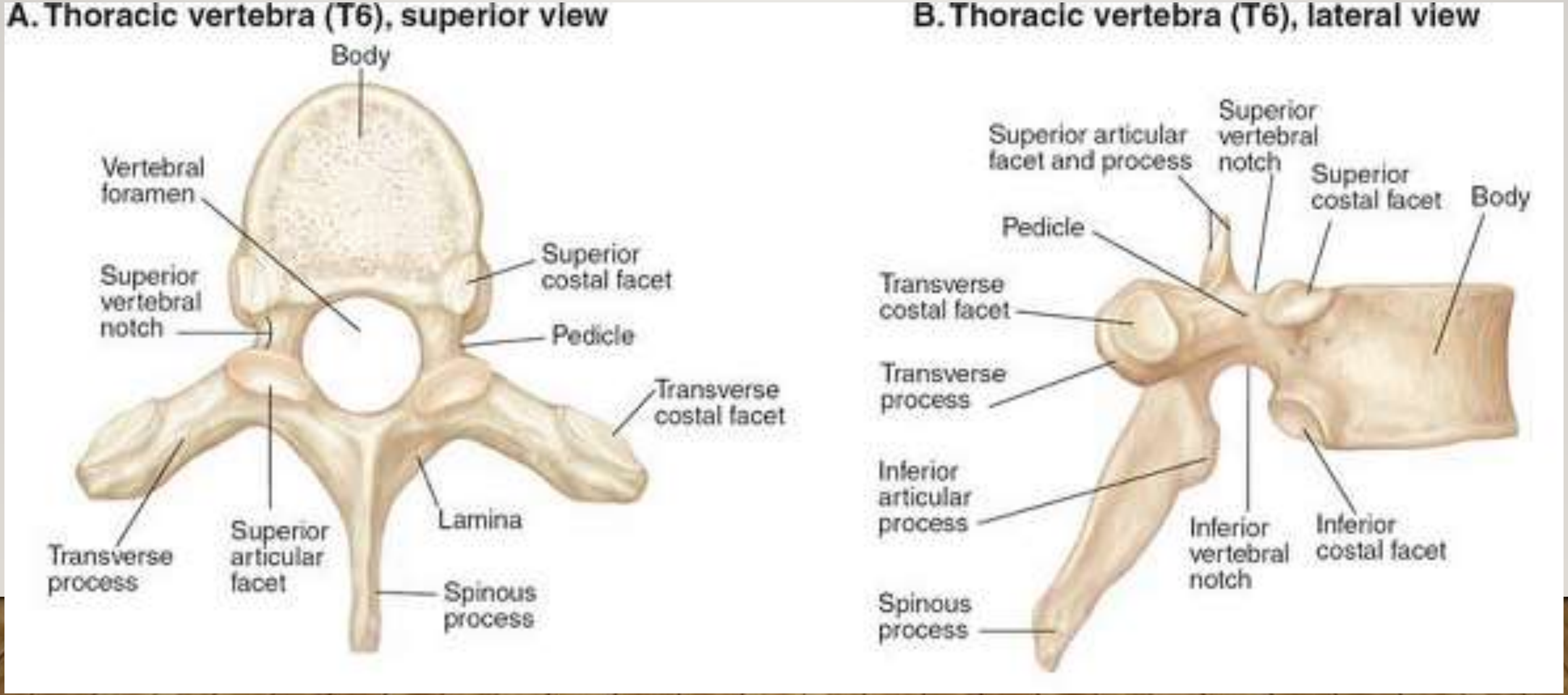

B
Atlas (CI vertebra)
Anterior tubercle
Facet for dens
Anterior arch
Lateral mass
Transverse process
Impressions for alar ligaments
Foramen transversarium
Facet for occipital condyle
Posterior arch
Posterior tubercle
Superior view
Transverse ligament of atlas
Superior view

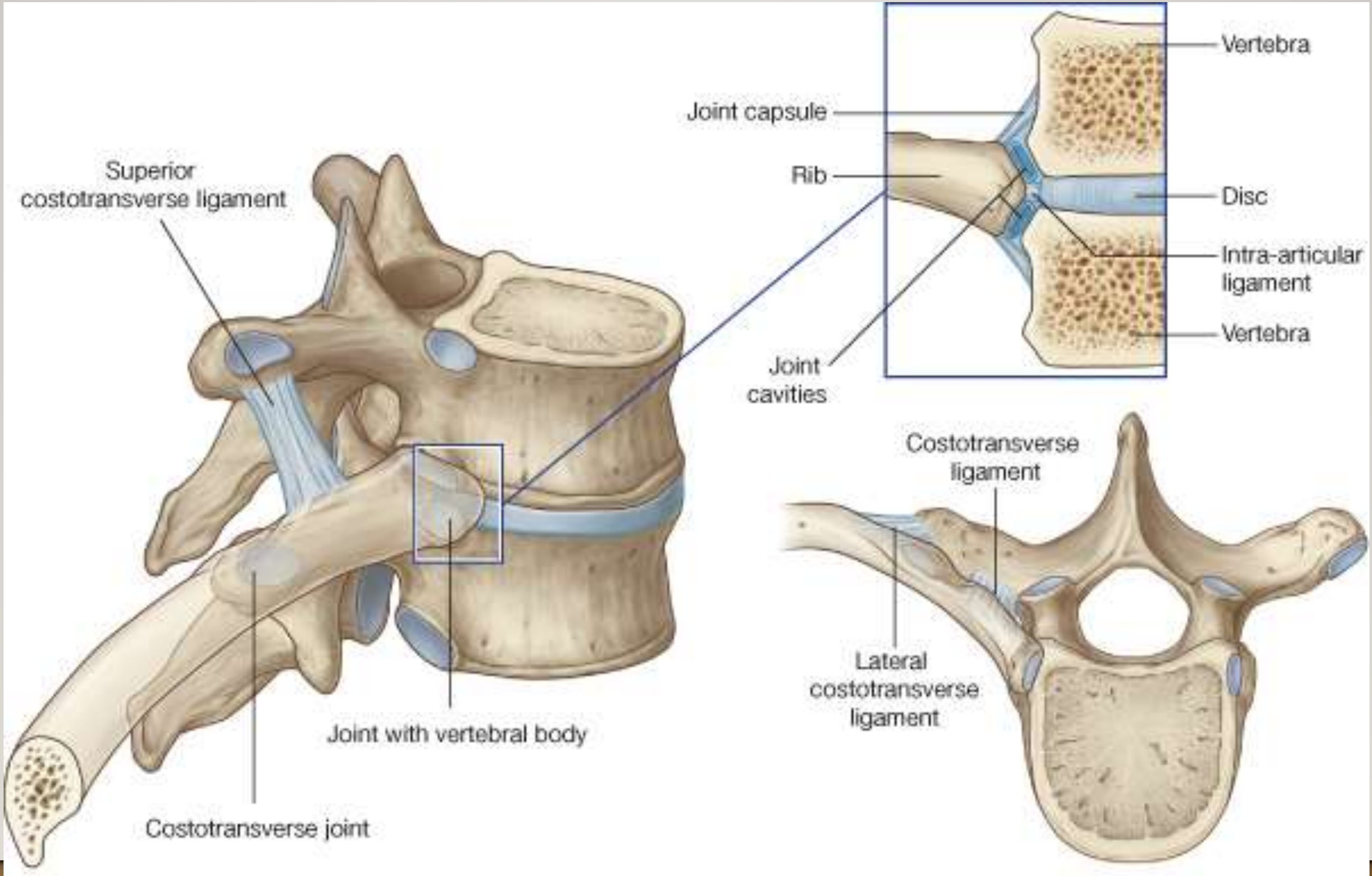
Axis


Dens
Axis (CII vertebra)
Transverse ligament of atlas
Alar ligaments
Dens
Superior view
Posterior view
Posterosuperior view

Thoracic Vertebrae:

- They are characterized by; no foramen transversarium, spine is long and directed downwards, an upper and lower costal facet is present on either side of the body and the transverse process near its end shows an articular facet.
- The first, 11th and 12th vertebrae are modified from the general characteristic pattern so they are atypical.

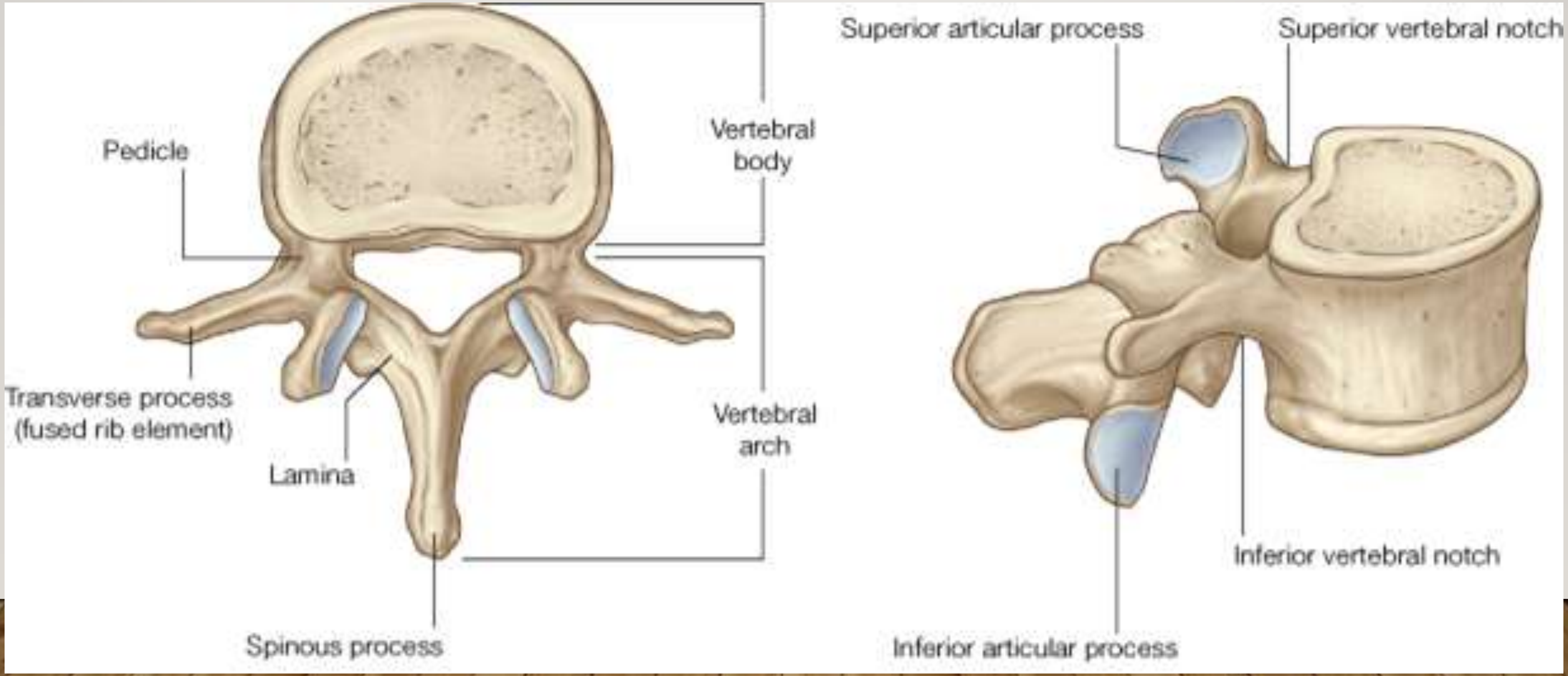


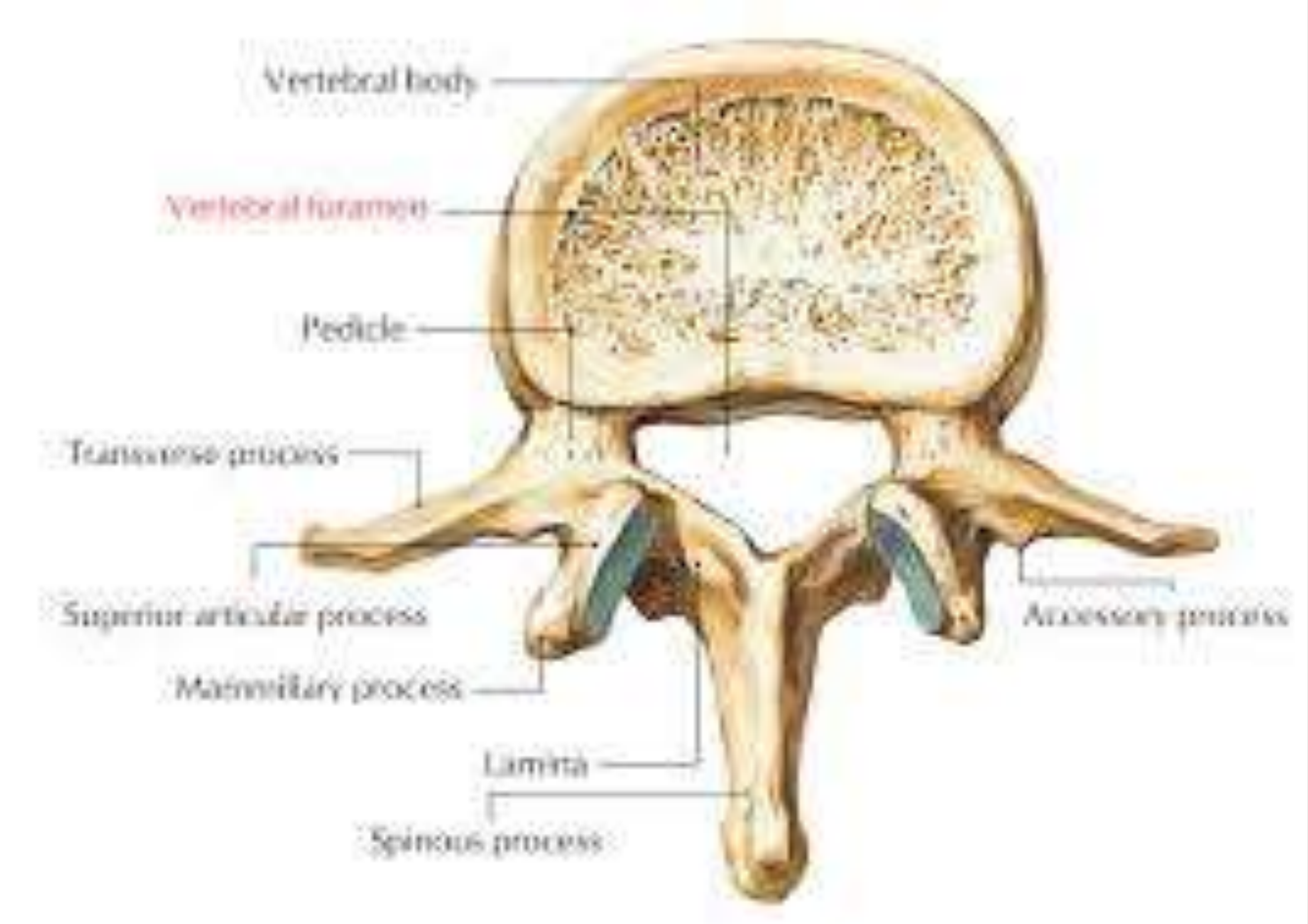


Superior
costotransverse ligament
Joint with vertebral body
Costotransverse joint
Joint capsule
Rib
Joint
cavities
Vertebra
Disc
Intra-articular
ligament
Vertebra
Costotransverse
ligament
Lateral
costotransverse
ligament

Lumbar Vertebrae:

- They are 5 in number.
- They are characterized by large size, and no costal facets.
- The transverse process is long, tapering with no foramen.
- There are mamillary and accessory processes.
- The spine is quadrangular and horizontal.

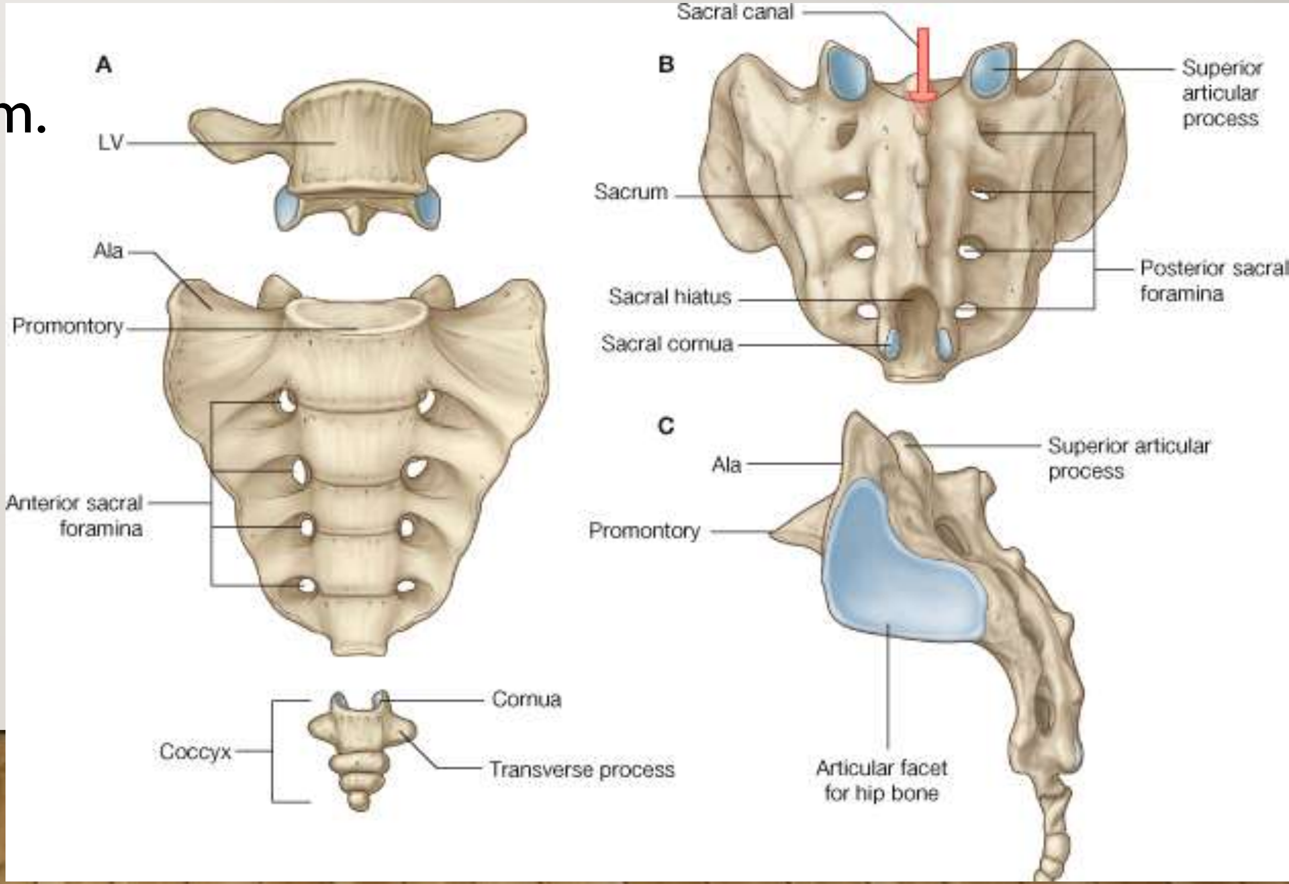



Vertebral body
Vertebral foramen
Pedicle
Transverse process
Superior articular process
Mammillary process
Lamina
Spinous process
Accessory process

THE SACRUM

- They are 5 vertebrae fused together forming one single bone.
- It is triangle in shape having apex at the lower end while the base is at its upper end.
- The sacrum forms the posterior part of the bony pelvis, wedged in between the right and left hip bones.
- **Articulations:**

1. The base articulates with the 5th lumbar vertebra.
2. The apex articulates with the coccyx.
3. The lateral surface articulates with the ileum.



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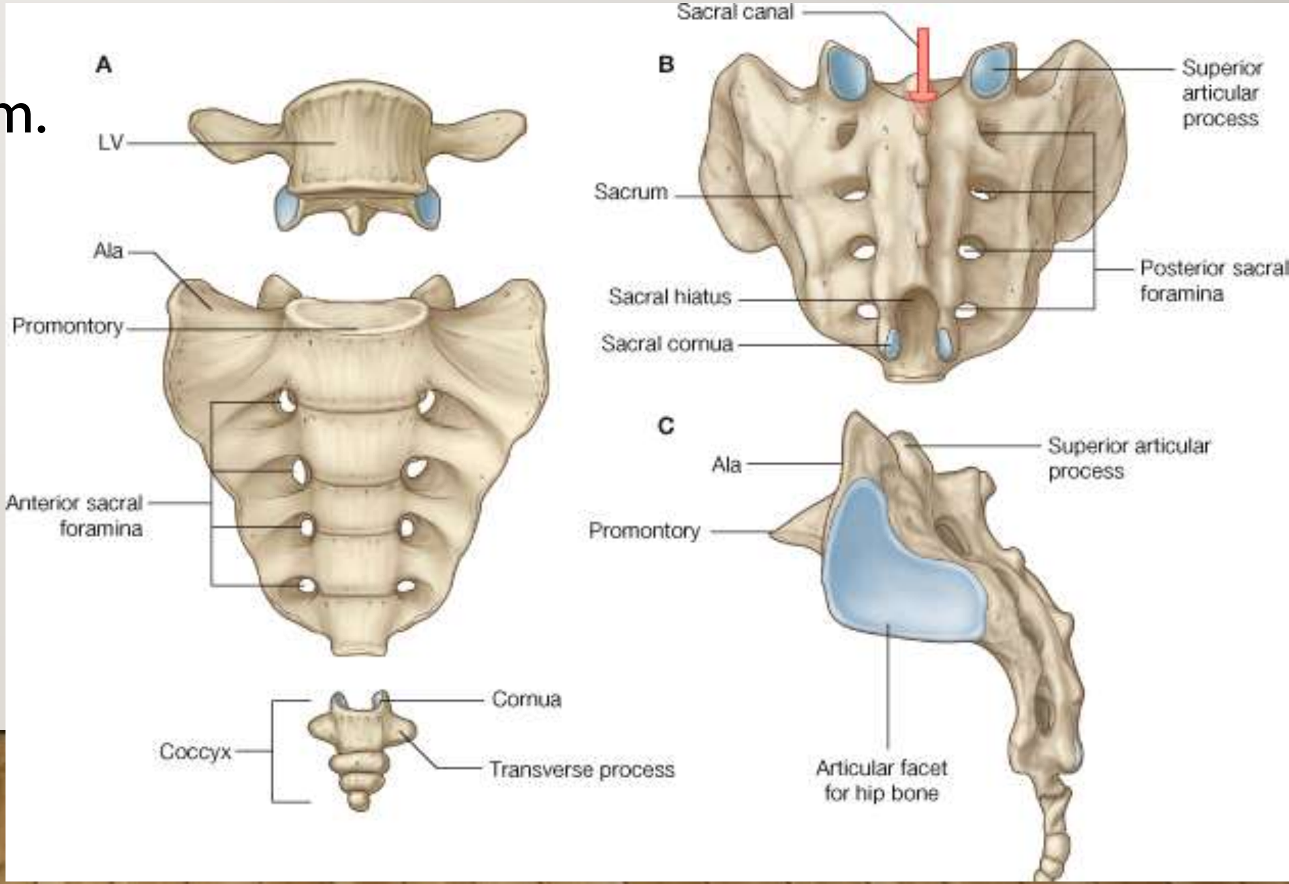
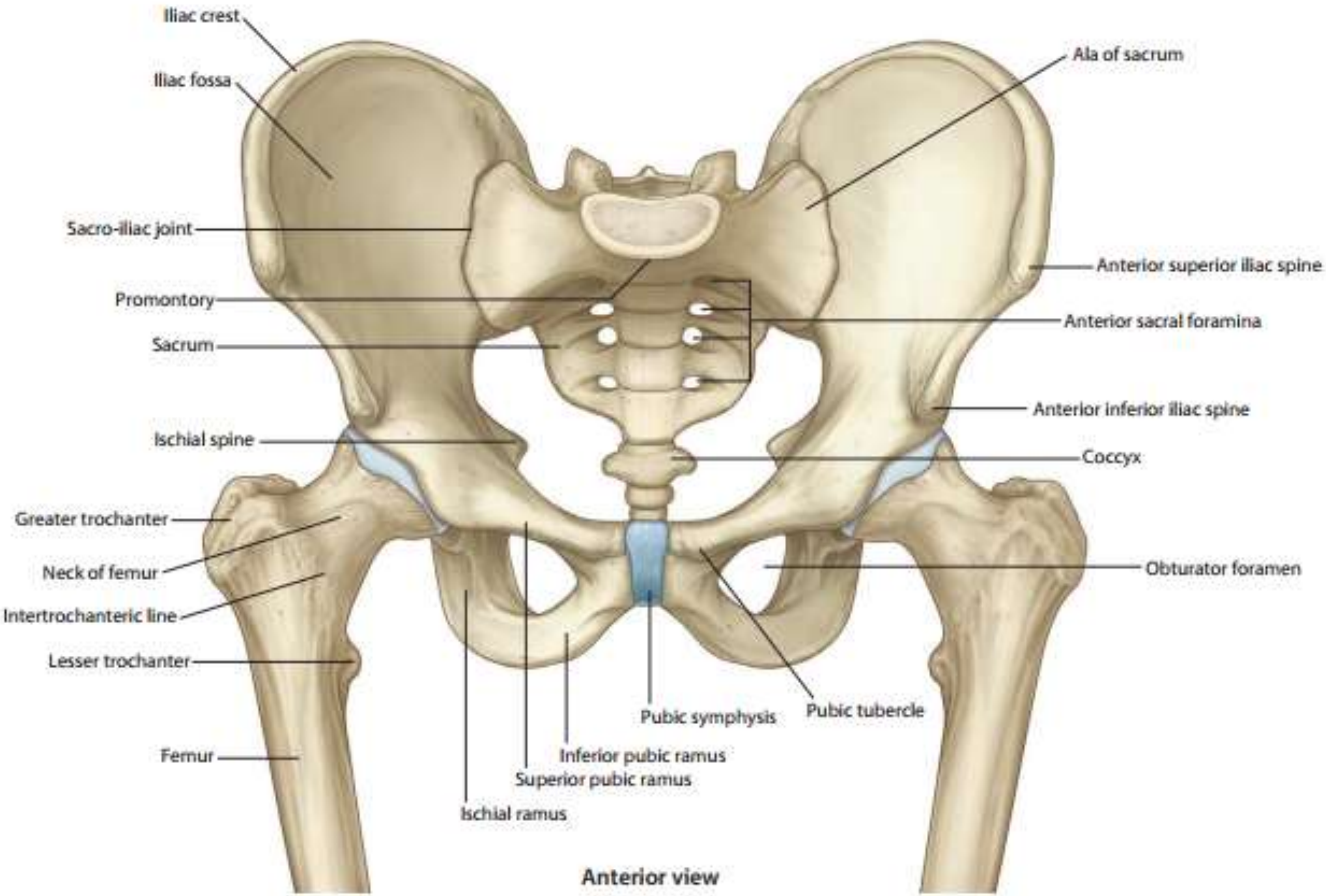




Anterior view

There are ….. Cervical vertebrae in human body
There are …..thoracic vertebrae in human body

<u>True or false</u>
Cervical vertebrae contains foramen transversorium

There are five fused lumbar vertebrae in human body

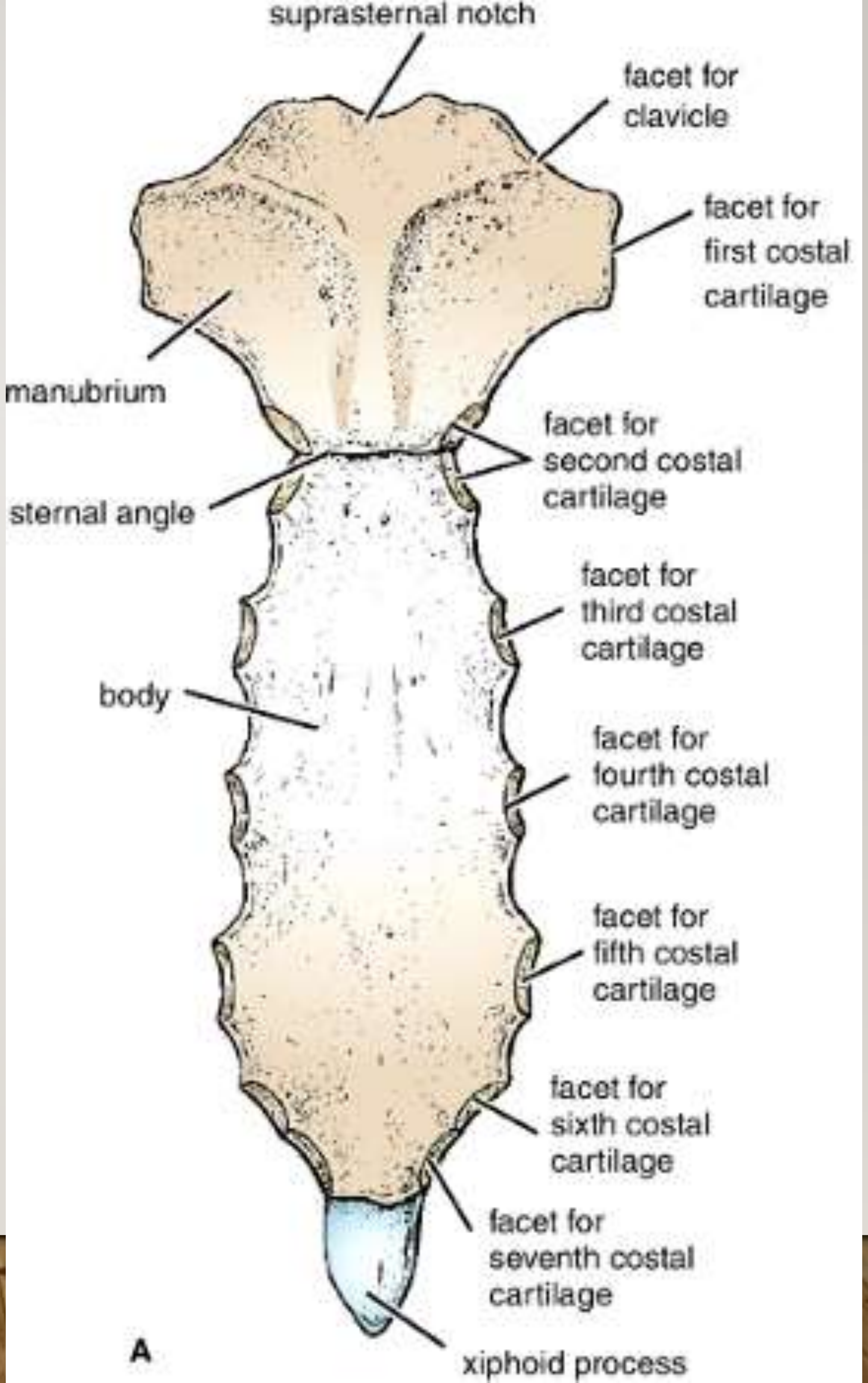
There are seven fused sacral vertebrae in human body

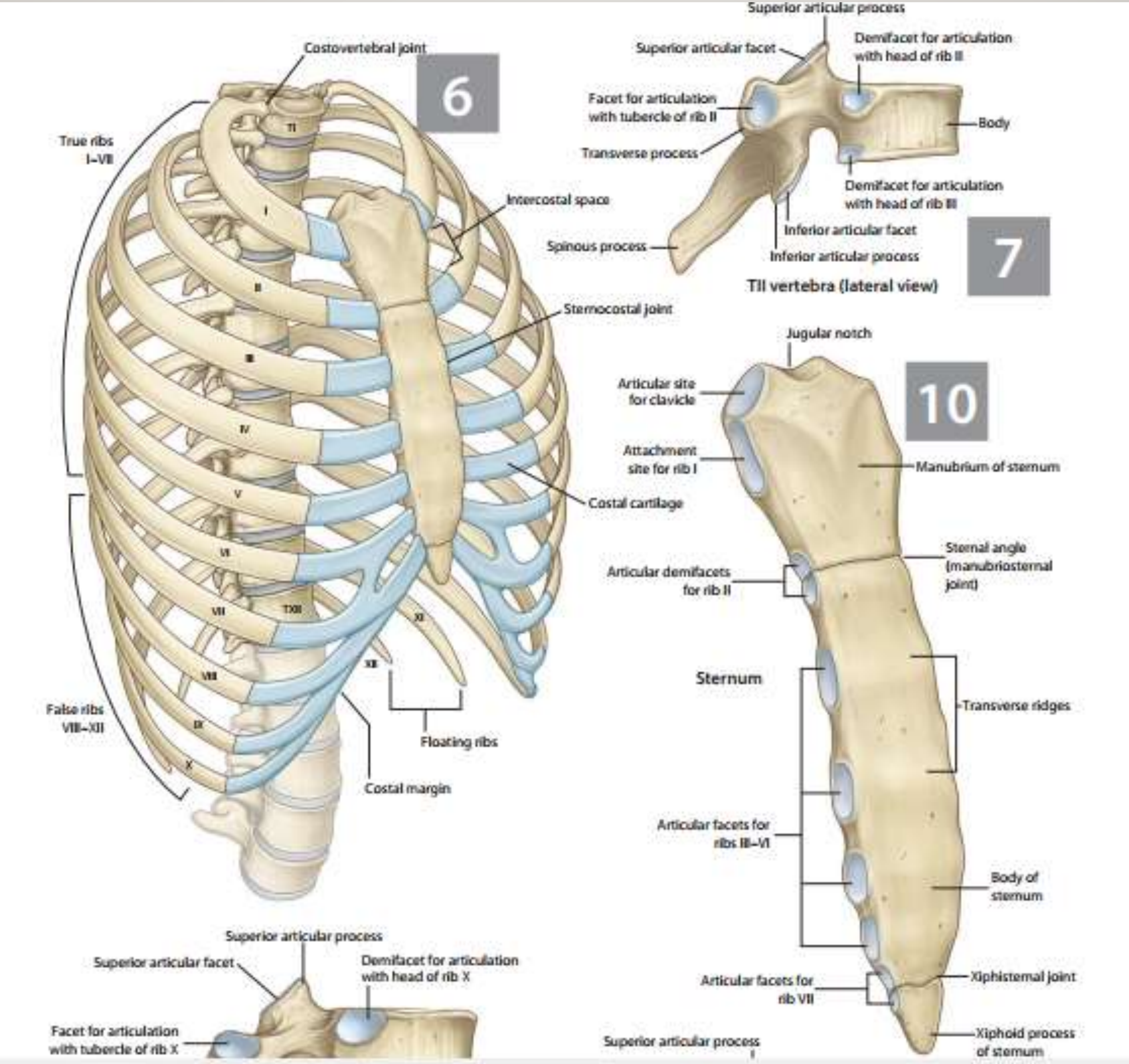
Thoracic vertebrae have costal facets for articulation with the ribs

STERNUM

- It is an elongated flat bone located ventrally in the midline of thorax.
- It is formed of manubrium, body and xiphoid process from above downwards.
- **Articulations:**

1. The manubrium sterni articulates above and laterally with the sternal end of the clavicle **(sternoclavicular joint)**.
2. The lateral border of the sternum articulates with the costal cartilages of the <u>upper seven ribs</u> **(sternocostal joint)**.

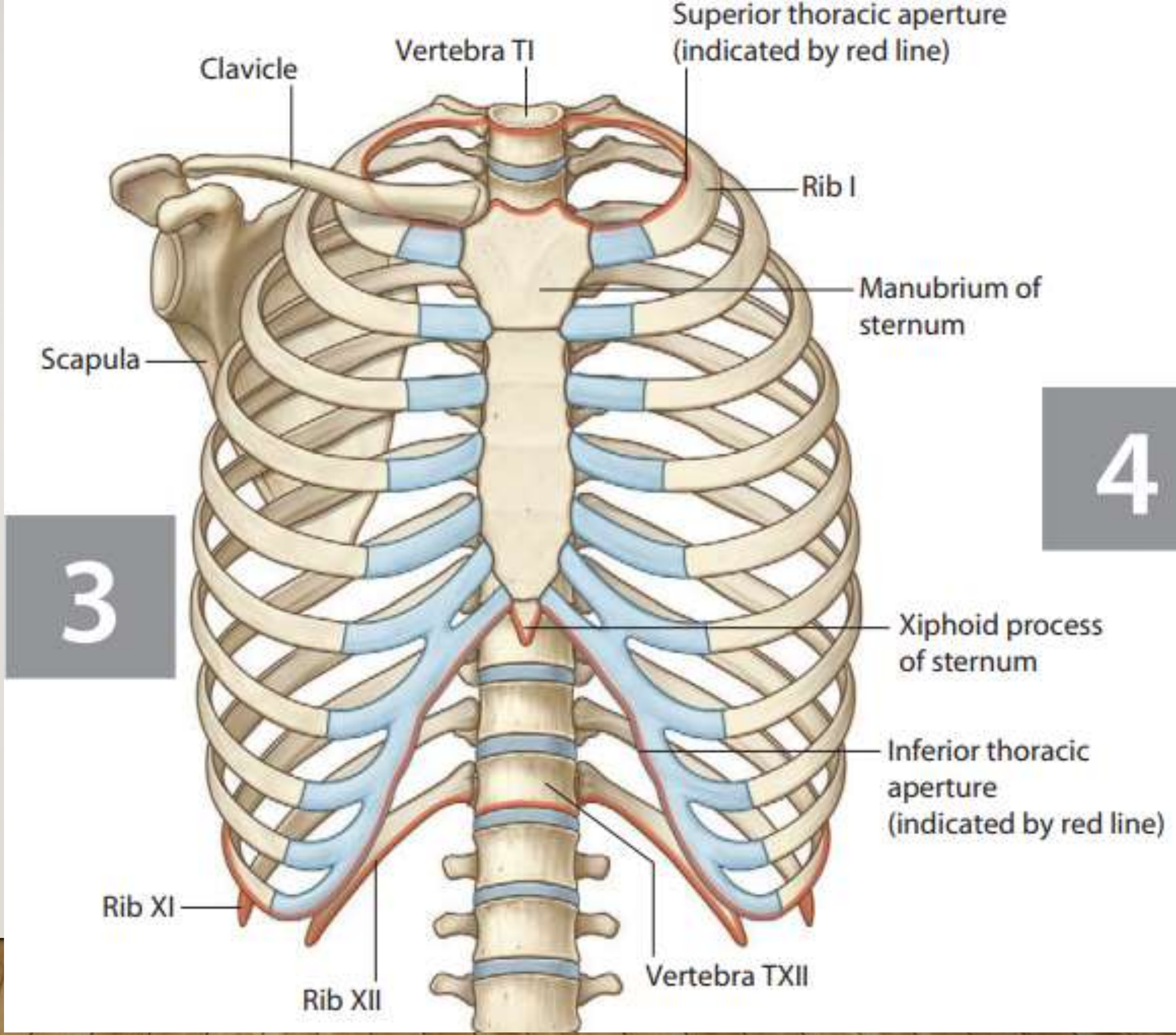



6
Costovertebral joint
True ribs I–VII
Intercostal space
Sternocostal joint
Costal cartilage
False ribs VIII–XII
Floating ribs
Costal margin
7
Superior articular process
Superior articular facet
Demifacet for articulation with head of rib II
Facet for articulation with tubercle of rib II
Body
Transverse process
Demifacet for articulation with head of rib III
Inferior articular facet
Spinous process
Inferior articular process
TII vertebra (lateral view)
10
Jugular notch
Articular site for clavicle
Attachment site for rib I
Manubrium of sternum
Sternal angle (manubriosternal joint)
Articular demifacets for rib II
Sternum
Transverse ridges
Articular facets for ribs III–VI
Body of sternum
Xiphisternal joint
Articular facets for rib VII
Xiphoid process of sternum
Superior articular process
Superior articular facet
Demifacet for articulation with head of rib X
Facet for articulation with tubercle of rib X

The Ribs

- They are 12 ribs on each side. It forms a series of obliquely placed arches that make up a large part of the thoracic cage.
- Types of ribs:

1. True: The upper seven ribs. They articulate anteriorly by their costal cartilages with the sternum.
2. False: The 8th, 9th and 10th, each of their costal cartilage joins the one above it.
3. Floating: 11th and 12th with free anterior end.

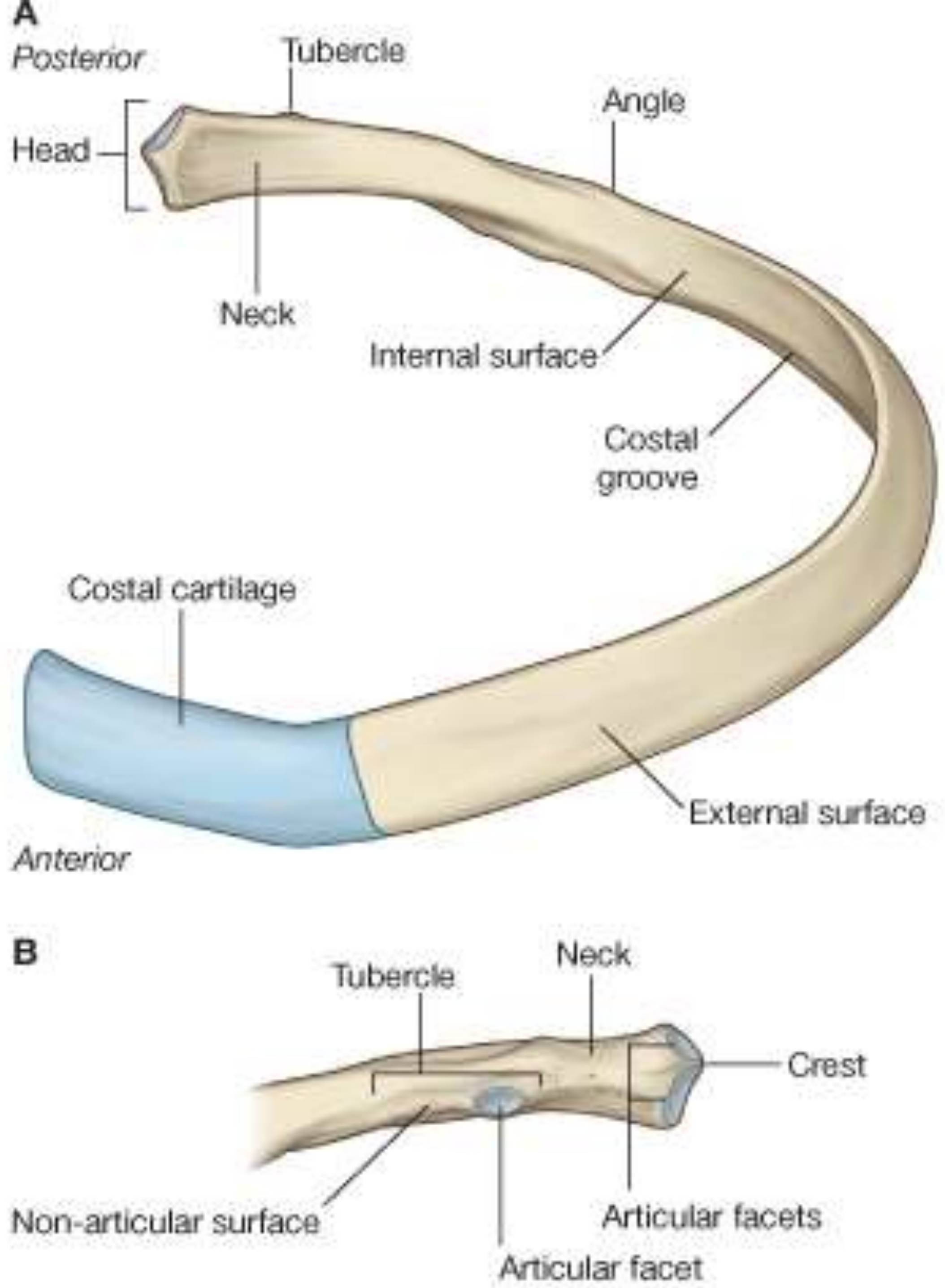


The Ribs

➢ Features of a typical rib:

The rib is composed of:

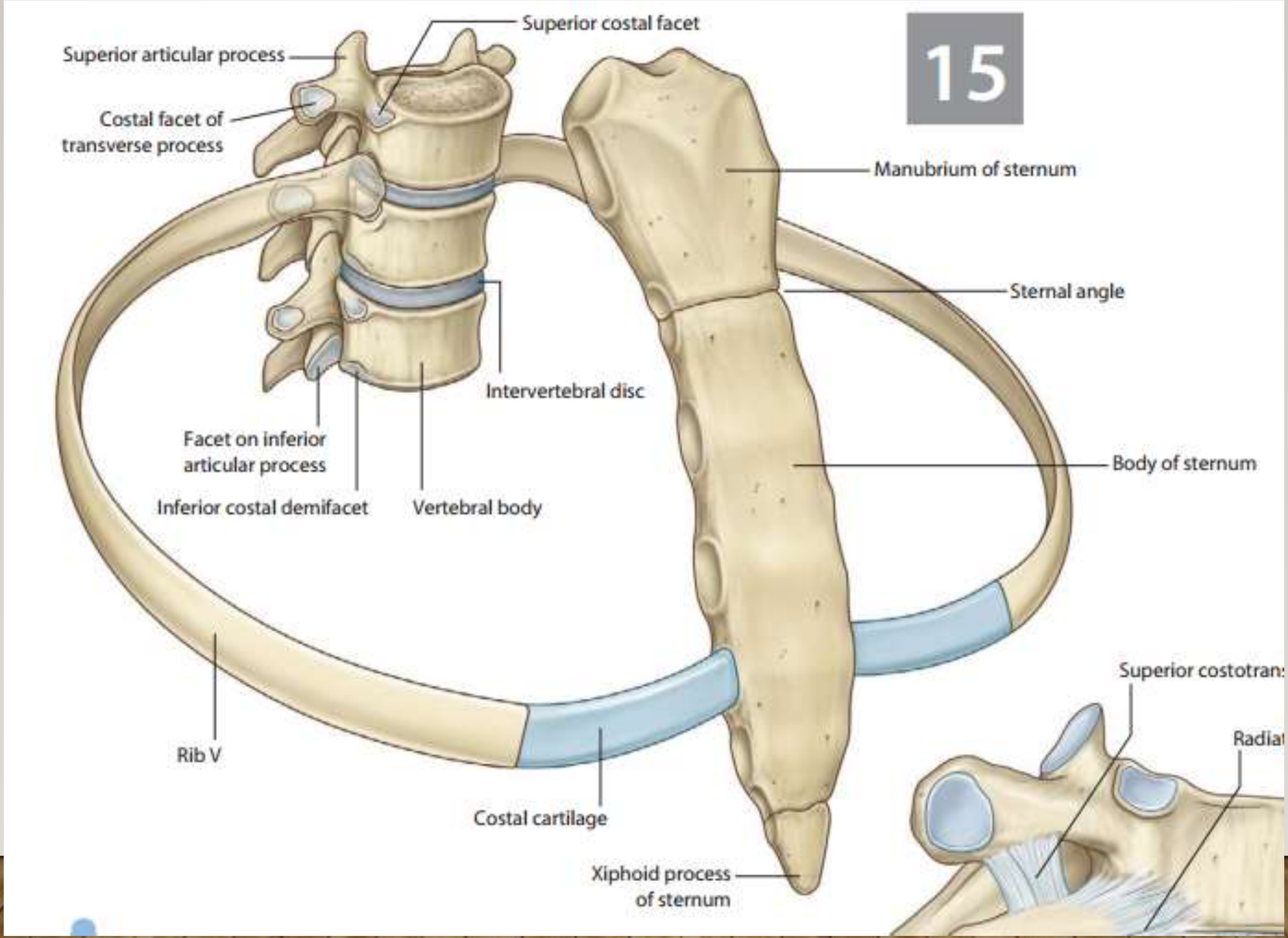
1. An anterior or sternal end attached to costal cartilage.
2. A posterior or vertebral end and is formed of a head, neck and a tubercle.
3. A shaft inbetween which is twisted antero-posteriorly.
4. Its upper border is rounded; its lower border is sharp.
5. The costal groove is near its lower border.



The Ribs

➢ **Articulations:**

1. Posteriorly: All ribs articulate by their heads with the bodies of thoracic vertebrae.
2. The upper ten ribs articulate by their tubercles with the transverse processes.
3. Anteriorly: The cartilages of the true ribs articulate with the sternum, those of the false ribs articulate together. The floating ribs are free.


15
Superior costal facet
Superior articular process
Costal facet of
transverse process
Manubrium of sternum
Sternal angle
Intervertebral disc
Facet on inferior
articular process
Inferior costal demifacet
Vertebral body
Body of sternum
Rib V
Costal cartilage
Xiphoid process
of sternum
Superior costotran
Radiat

The true ribs articulates through their heads with ………

True or false

There are 7 pairs of true ribs that articulates with the sternum through their costal cartilages

11th and 12 th ribs are called floating ribs


Thank
you